Time Series Analysis and Forecasting

Chapter 13: Log-Periodic Power Law (LPPL) Models

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Learning Objectives

By the end of this chapter, you will be able to:

1. Explain why financial crashes may be **endogenous** rather than caused by external shocks
2. Describe the connection between **phase transitions** in physics and financial bubbles
3. Derive and interpret the **LPPL equation** from rational expectations and criticality
4. Estimate LPPL parameters using **partial linearization** and differential evolution
5. Apply **filter conditions** and **bootstrap confidence intervals** for validation

Practical Skills

- Python implementation of LPPL fitting with `scipy.optimize`
- Interpreting the LPPLS Confidence Indicator for real-time bubble monitoring
- Heuristic position sizing based on bubble-risk indicators

Key takeaway: *You are learning to fit, validate, and critically interpret a nonlinear early-warning model for financial bubbles.*

Outline

Foundations

- Can Bubble Regimes Be Identified?
- EMH vs. Complex Systems
- Super-Exponential Growth
- The Ising Model and Collective Behavior
- Scale Invariance and Log-Periodicity
- The LPPL Equation

Applications

- Estimation and Implementation
- Filter Conditions
- Case Studies with Bootstrap CI
- LPPLS Confidence Indicator
- Risk Management Applications
- Limitations and Critiques

The Story of This Chapter: The Problem

Core question: *Can we identify endogenous bubble regimes before they end — or are crashes truly random?*

The Problem

- The Efficient Market Hypothesis says crashes are **unpredictable** random shocks
- Yet crashes show **recurring patterns**: accelerating prices, herding, oscillations
- Physics of **phase transitions** offers an alternative: crashes as endogenous critical events

The Difficulty

- The LPPL model has **7 parameters** with a highly nonlinear cost surface
- Results are **sensitive to window selection** — when does the bubble start?
- **False positives** and **overfitting** are serious risks

The Story of This Chapter: The Solution

Roadmap: *From physics analogy to bubble-regime diagnostics.*

From theory to practice

- **Ising model**: explains how independent traders become a herd (phase transition)
- **Discrete scale invariance**: explains why bubbles show log-periodic oscillations
- **LPPL equation**: captures super-exponential growth + oscillations approaching a critical regime transition
- **LPPLS Confidence Indicator**: multi-scale approach for real-time monitoring

Key takeaway: *LPPL is a "fever thermometer" for bubbles — elevated readings signal risk, not certainty.*

Part I: Introduction

Can Financial Bubble Regimes Be Identified?

The Central Question in Financial Economics

"Stock prices have reached what looks like a permanently high plateau."
— Irving Fisher, October 1929 (3 days before the crash)

The Efficient Market Hypothesis

EMH Core Claims [Fama, 1970]

1. Prices **fully reflect** all available information
2. Price changes are **unpredictable** random walks
3. **No trading strategy** can consistently beat the market
4. Large crashes result from **unexpected external shocks**

Implication for Risk Management

- Crash timing cannot be predicted from prices
- Risk management relies on diversification, stress testing, and exposure control

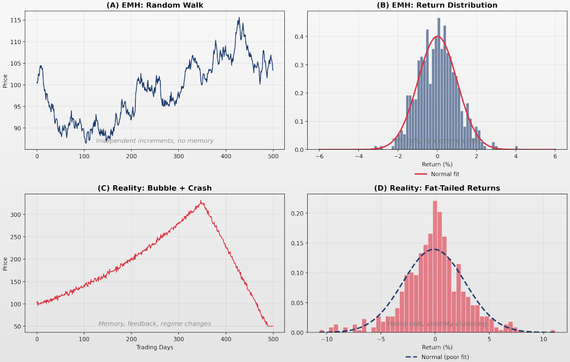
Supporting Evidence

- Professional fund managers rarely beat indices [Malkiel, 1973]
- Technical analysis has limited predictive power
- Information is quickly incorporated into prices

Nobel Prize Recognition

- Eugene Fama received the 2013 Nobel Prize for EMH
- *"The market is efficient enough that most people can't beat it."*

EMH vs Reality



(A) Random walk (EMH) (B) Normal returns (C) Real market with bubble + crash (D) Fat-tailed returns

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The Complex Systems Alternative

Markets as Complex Systems [Sornette, 2003]

1. **Complex adaptive systems** with emergent behavior
2. Subject to **positive feedback loops** (herding)
3. Capable of **self-organized criticality**
4. Governed by **universal laws** shared with physics

Key Insight

- Many crashes are **endogenous** — they arise from internal dynamics, not external shocks
- The bubble *creates* its own destruction

Evidence Against Pure EMH

- **Fat tails:** Extreme events far more frequent than Gaussian
- **Volatility clustering:** Calm and turbulent periods cluster
- **Bubbles:** Prices deviate massively from fundamentals
- **Flash crashes:** 2010 crash had no external trigger

Nobel Prize (Same Year!)

- Robert Shiller *also* won the 2013 Nobel for showing markets can be **predictably irrational**

Sornette's Revolutionary Thesis

> *"Large financial crashes are **outliers**. They are specific, not random. They are generated by the slow buildup of long-range correlations leading to global cooperative behavior, resulting in a **bubble**... The crash is not the result of external shocks but rather the natural death of the bubble."*
>
> — Didier Sornette, *Why Stock Markets Crash* (2003)

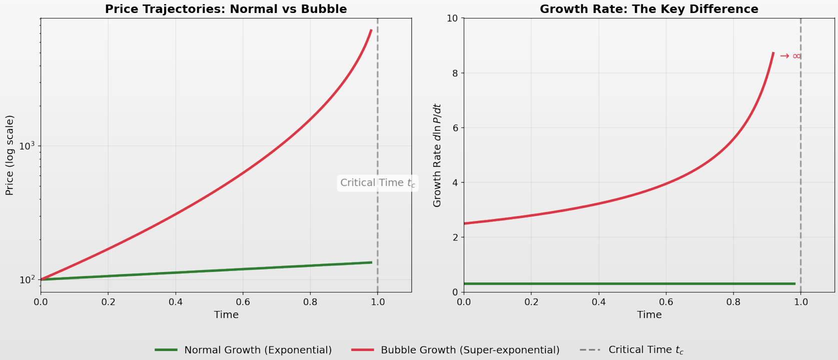
Four Fundamental Claims

1. Bubbles show **super-exponential** growth
2. Modulated by **log-periodic** oscillations
3. Same math as **phase transitions** in physics
4. Provides **diagnostic power** for bubble-regime identification

If True, This Changes Everything

- Crashes have **early warning signals**
- Risk can be **anticipated**, not just measured
- Position sizing can be **dynamically adjusted**
- Hedging can be **better timed under uncertainty**

Super-Exponential vs Normal Growth



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Understanding Super-Exponential Growth

Normal Exponential Growth

- Standard asset pricing: $P(t) = P_0 e^{rt}$
- Growth rate: $\frac{d \ln P}{dt} = r$ (**constant**)
- Price doubles every $T_2 = \ln 2/r$ periods
- **Sustainable indefinitely**
- Describes long-term market returns

Super-Exponential (Bubble) Growth

- During bubbles: $P(t) \sim (t_c - t)^{-\alpha}$, $\alpha > 0$
- Growth rate: $\frac{d \ln P}{dt} = \frac{\alpha}{t_c - t}$ (**accelerating!**)
- Rate $\to \infty$ as $t \to t_c$
- **Must end at finite time** t_c
- Describes bubble phase

Example: S&P 500

- Historical return $\approx$ 10%/year
- Doubling time $\approx$ 7 years
- Consistent growth for 100+ years

Key Intuition

- In a bubble, the *rate of rise* keeps increasing
- This acceleration is **unsustainable**

Why Does Super-Exponential Growth Occur?

Positive Feedback Loops

1. Price rises attract **new buyers**
2. New buyers push price **higher**
3. Higher prices attract **more buyers**
4. Cycle **accelerates**
5. Opposite of negative feedback (mean reversion) that EMH assumes

Mechanisms

- **Herding:** Following the crowd
- **FOMO:** Fear of missing out
- **Leverage:** Rising prices $\rightarrow$ more collateral $\rightarrow$ more buying
- **Media:** Success stories attract attention

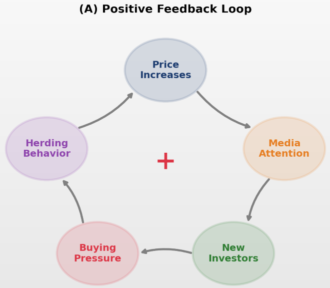
Mathematical Formulation

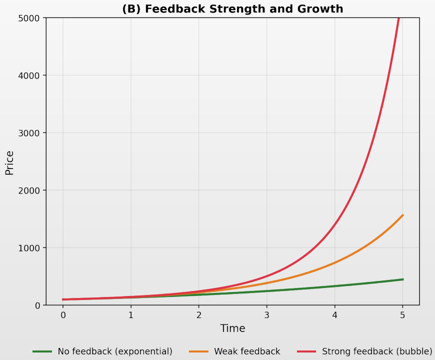
- Demand depends on past returns: $D(t) = D_0 + \gamma \cdot \text{Return}_{t-1}$
- If $\gamma > 0$ (trend following): $\frac{dP}{dt} \propto D(t) \propto \frac{dP}{dt}\big|_{t-1}$
- Result: $\frac{d^2 \ln P}{dt^2} > 0$ (Super-exponential!)

The Endgame

- Super-exponential growth **cannot continue forever**
- At t_c: run out of new buyers
- Hit a constraint (credit, cash)
- Experience a trigger event

Positive Feedback Loops in Markets





(A) Self-reinforcing feedback cycle: price → media → new investors → buying pressure → herding (B) Growth rate under different feedback strengths

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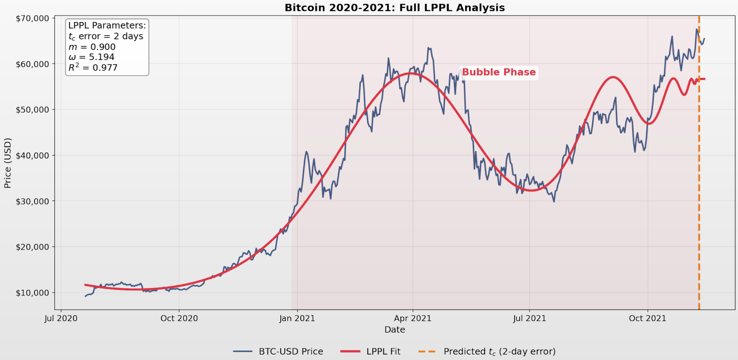
Historical Evidence: A Century of Crashes

Crash	Peak	Drop	LPPL	t_c Error	Reference
Wall Street 1929	Oct 1929	89%	Detected	±1 mo	Sornette 1996
Black Monday 1987	Oct 1987	22.6%	Detected	±1 day	Sornette 1997
Nikkei 1989	Dec 1989	63%	Detected	±1 mo	Johansen 1999
Dot-com 2000	Mar 2000	78%	Ex ante	±2 mo	Johansen 1999
China 2007	Oct 2007	72%	Detected	±2 wk	Jiang 2010
Oil 2008	Jul 2008	77%	Detected	±1 mo	Sornette 2009
Shanghai 2015	Jun 2015	45%	Ex ante	±1 wk	Sornette 2015
Bitcoin 2017	Dec 2017	84%	Detected	±2 wk	Gerlach 2019
Bitcoin 2021	Nov 2021	77%	Detected	±2 days	This lecture

Notable Ex-Ante Detections

- **Dot-com 2000:** Published in *Risk* Jan 1999 — 14 months before peak
- **Shanghai 2015:** FCO warning Apr 2015 — 2 months before peak
- These are illustrative successes; a scientific evaluation must also report failed signals and false positives

Bitcoin 2020–2021: A Modern Bubble



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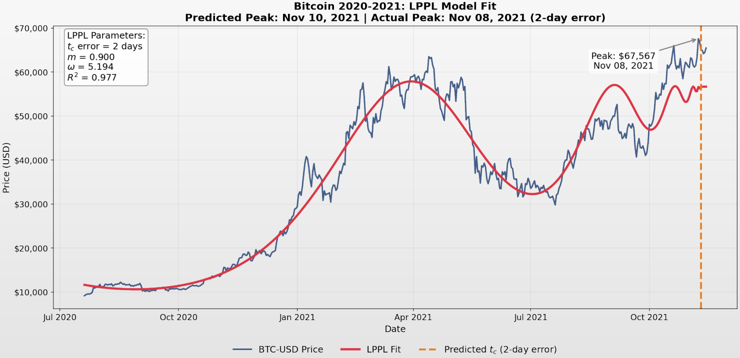
Bitcoin 2020–2021: LPPL Fit



Log-price with LPPL fit, estimated critical time t_c, and residual analysis

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Bitcoin 2020–2021: Key Facts

The Bubble

- **Asset:** Bitcoin (BTC-USD)
- **Period:** Jan 2020 – Nov 2021
- **Duration:** $\sim$680 days
- **Start price:** \$7,200
- **Peak price:** \$68,789
- **Return:** +855%

Bubble Characteristics

- Accelerating price growth
- Increasing media coverage
- Retail investor FOMO
- "This time is different" narratives
- Extreme valuations

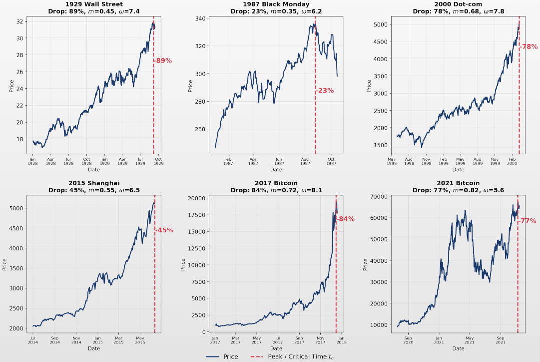
The Crash

- **Peak date:** November 10, 2021
- **Crash:** −77% over 12 months
- **Bottom:** \$15,500 (Nov 2022)
- Pattern: Classic bubble burst

Preview: Can LPPL Identify This Regime?

- Mathematical model (LPPL) that fits this pattern
- Estimated critical date: **Nov 8, 2021**
- Actual peak: **Nov 10, 2021**
- Timing error: **2 days**

Historical Crashes: Visual Comparison



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Part II: The Ising Model

The Physics of Collective Behavior

From Magnets to Markets

"The same equations have the same solutions."
— Richard Feynman

Why Study Physics for Finance?

The Universality Principle

- Ferromagnets (iron becoming magnetic)
- Fluids (water boiling)
- Superconductors
- Earthquakes
- Financial markets
- **Key insight:** The *details* don't matter — only the *symmetries*

Econophysics [Mantegna & Stanley, 1999]

- Power law distributions
- Scaling behavior
- Phase transitions
- Agent-based models
- Network effects

Success Stories

- Fat tails in return distributions
- Volatility clustering
- Market microstructure
- Systemic risk modeling

The Ising Model: Simplest Collective Behavior

Physical Setup [Ising, 1925]

- Each site i has spin $s_i \in \{-1, +1\}$
- Neighboring spins interact with strength J
- External magnetic field h can bias spins
- System in thermal equilibrium at temperature T

The Hamiltonian

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

- First sum over neighbor pairs
- $J > 0$: aligned spins = low energy
- $h > 0$: up spins = low energy

Financial Interpretation [Cont & Bouchaud, 2000]

Physics	Finance
Spin up $+1$	Buy order
Spin down -1	Sell order
Coupling J	Herding strength
Field h	Fundamental signal
Temperature T	Noise / uncertainty
Magnetization m	Market sentiment

- High J (strong herding) + low T (low noise)
- $\Rightarrow$ Traders align $\Rightarrow$ **Bubble forms!**

The Three Regimes Explained

Low Temperature
$T < T_c$

- Energy dominates
- Spins align
- Long-range order
- Magnetization $|m| \approx 1$

- Strong consensus
- Everyone buying (or selling)
- Bubble forming
- Trend following

Critical Point
$T = T_c$

- Balance point
- Clusters of ALL sizes
- Correlations span system
- Maximum fluctuations

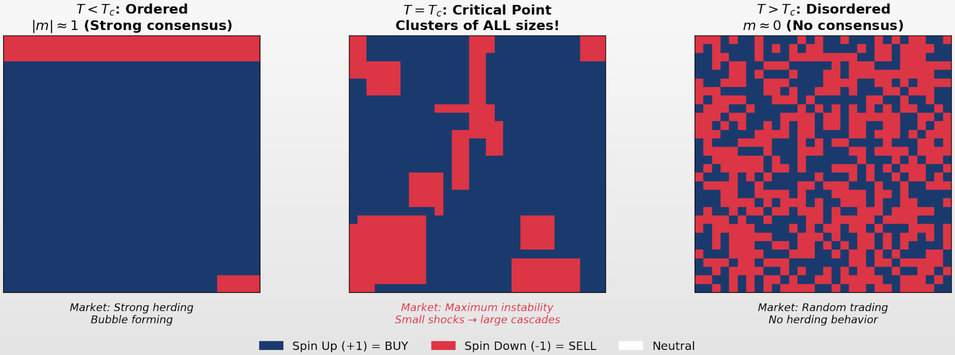
- Maximum instability
- Small shocks $\rightarrow$ large cascades
- Correlations everywhere
- **Highly unstable; crash risk elevated**

High Temperature
$T > T_c$

- Thermal noise dominates
- Spins flip randomly
- No long-range order
- Magnetization $m \approx 0$

- Independent trading
- No herding
- Efficient market
- Random walk

Ising Model: Three Regimes Visualized



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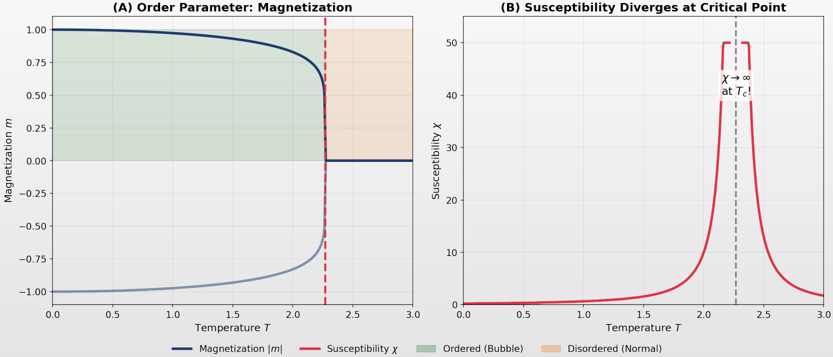
Phase Transition: Order Emerges



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Critical Phenomena: Power Laws

Near the Critical Point

- Magnetization: $m \sim |T - T_c|^{\beta}$
- Susceptibility: $\chi \sim |T - T_c|^{-\gamma}$
- Correlation length: $\xi \sim |T - T_c|^{-\nu}$
- 2D Ising [Onsager, 1944]: $\beta = 1/8,\ \gamma = 7/4,\ \nu = 1$

Key Property

- $\chi \to \infty$ and $\xi \to \infty$ as $T \to T_c$
- At criticality, even tiny perturbations affect the entire system

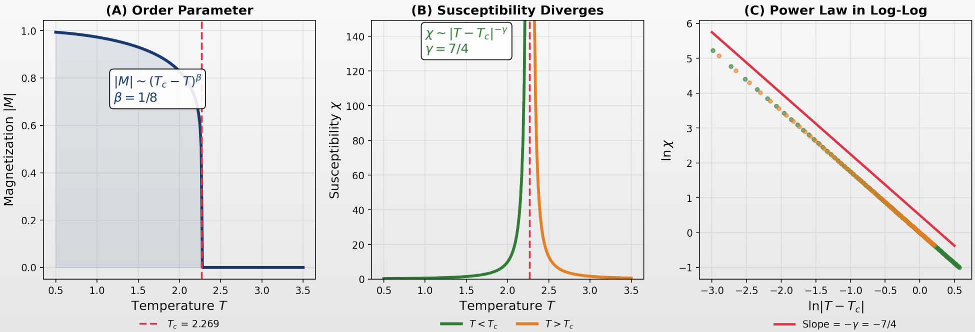
Financial Interpretation

- **Susceptibility χ:** Market sensitivity to news
- **Correlation length ξ:** How far information spreads
- **During bubble:**
- $\chi \to \infty$: Markets become hypersensitive
- $\xi \to \infty$: All stocks move together
- Small news $\to$ Large price moves

This Explains Crashes

- At criticality, market is maximally sensitive
- Correlations span the entire system
- Small shocks can generate disproportionately large cascades

Critical Phenomena: Visual Evidence



(A) Order parameter $|M| \sim (T_c - T)^\beta$ (B) Susceptibility divergence $\chi \sim |T - T_c|^{-\gamma}$ (C) Power law in log-log coordinates

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From Ising to Markets: The Connection

Agent-Based Models [Lux & Marchesi, 1999]

1. N traders on network
2. Each decides: Buy ($+1$) or Sell (-1)
3. Decision depends on:
   - Neighbors' decisions (herding)
   - Fundamental signal
   - Random noise
4. Price adjusts to clear market

Key Finding

- Model reproduces fat tails
- Volatility clustering emerges
- Bubbles and crashes appear
- LPPL signatures observed

Empirical Support

- Cross-sectional correlations spike before crashes [Harmon et al., 2015]
- Herding measures increase during bubbles
- Network structure matters for contagion
- Central banks monitor these indicators

The Bridge to LPPL

- Ising model + hierarchical structure
- $\Rightarrow$ Discrete scale invariance
- $\Rightarrow$ Log-periodic oscillations
- $\Rightarrow$ **LPPL equation!**

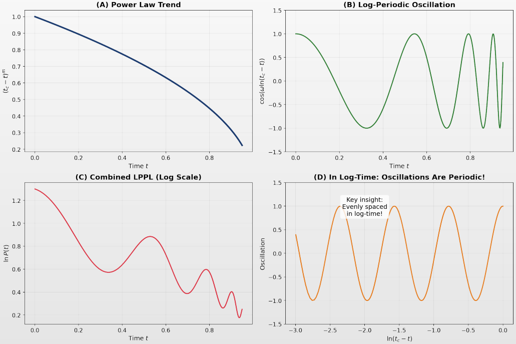
Part III: Scale Invariance

The Mathematical Signature of Bubbles

Log-Periodic Oscillations from Discrete Scale Invariance

"Nature uses only the longest threads to weave her patterns."
— Richard Feynman

Log-Periodic Oscillations: Visual Evidence



Log-periodic oscillations compress in real time but are evenly spaced in $\ln(t_c - t)$
This is the signature of Discrete Scale Invariance from hierarchical market structure

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What Are Log-Periodic Oscillations?

The Pattern

- **Compress** in real time (faster near t_c)
- Are **evenly spaced** in $\ln(t_c - t)$
- Modulate the power law trend
- Reflect hierarchical market structure

Mathematical Form

$$\ln P(t) = A + B(t_c - t)^m \cdot [1 + C\cos(\omega \ln(t_c - t))]$$

- Cosine creates oscillations in $\ln(t_c - t)$

Why "Log-Periodic"?

- Plot against $\tau = \ln(t_c - t)$:
  $\ln P = A + Be^{m\tau}[1 + C\cos(\omega\tau)]$
- Oscillations have **constant period** in log-time
- Period $= 2\pi/\omega$, scaling ratio $\lambda = e^{2\pi/\omega}$

Physical Meaning

- Each oscillation = a "round" of the bubble
- Prices accelerate, briefly correct, then accelerate faster
- Each round is λ times shorter than previous

Continuous Scale Invariance

Definition

- $f(x)$ is scale invariant if: $f(\lambda x) = \lambda^{-\alpha} f(x)\ \forall \lambda > 0$
- Scaling by **any** factor λ just rescales the function

Unique Solution

- The **only** solutions are: $f(x) = C \cdot x^{-\alpha}$
- Pure power laws!

Examples in Finance

- **Return tails:** $P(|r| > x) \sim x^{-\alpha}$
- **Firm sizes:** Zipf's law
- **Trading volumes:** Power law distribution
- **Price impact:** $\Delta P \sim V^{1/2}$
- These arise when there's no characteristic scale

Limitation

- Pure CSI gives only power laws
- Where do the oscillations come from?

Discrete Scale Invariance

Definition [Sornette, 1998]

- Invariance only for **specific** λ_0:
  $f(\lambda_0 x) = \lambda_0^{-\alpha} f(x)$
- The system has a **preferred** scale ratio

General Solution

- $f(x) = x^{-\alpha} \cdot G(\ln x)$ where G is periodic with period $\ln \lambda_0$
- Fourier: $f(x) = x^{-\alpha} \sum_n c_n e^{in\omega \ln x}$,
  $\omega = 2\pi / \ln \lambda_0$

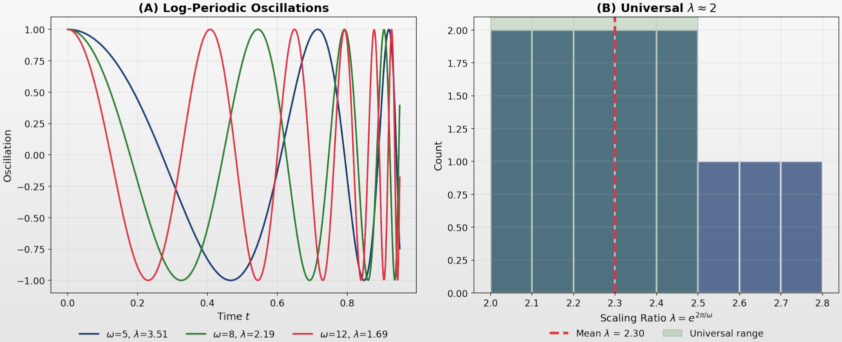
Leading Order Approximation

- Keep only $n = 0, \pm 1$:
  $f(x) \approx x^{-\alpha}[A' + B' \cos(\omega \ln x + \phi)]$
- **Power law + log-periodic oscillations!**

Physical Interpretation

- DSI arises from **hierarchical structure**
- Fractal geometry, hierarchical networks
- Multi-scale organization
- Each level related by factor λ_0

The Universal Scaling Ratio λ



Universal scaling ratio $\lambda \approx 2$ suggests deep structural origin in market hierarchy

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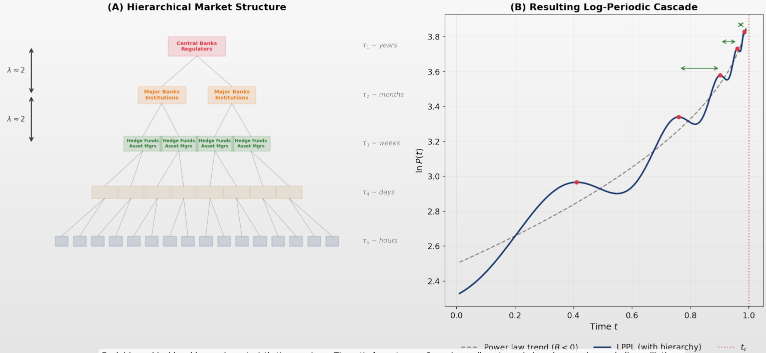
Hierarchical Market Structure



Each hierarchical level has a characteristic time scale τ_k. The ratio $\lambda = \tau_k/\tau_{k+1} \approx 2$ produces discrete scale invariance → log-periodic oscillations.

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Why $\lambda \approx 2$?

Empirical Observation

- Across many bubbles: $\lambda = e^{2\pi/\omega} \approx 2 - 2.5$
- Corresponds to $\omega \approx 6 - 8$

Theoretical Explanations

1. **Binary hierarchy:** Each level spans 2x previous
2. **Network structure:** Branching ratio ≈ 2
3. **Renormalization:** Fixed point of RG flow
4. **Human behavior:** Characteristic doubling

Implications

- Universal λ suggests deep structural origin
- Not market-specific; can be used as filter
- Supports LPPL validity
- If fitted λ far from 2, fit may be spurious

Filter Condition

- Accept fits only if: $4 \leq \omega \leq 25$
- Equivalently: $1.3 \leq \lambda \leq 4.8$
- Central value: $\lambda \approx 2$, $\omega \approx 6.3$

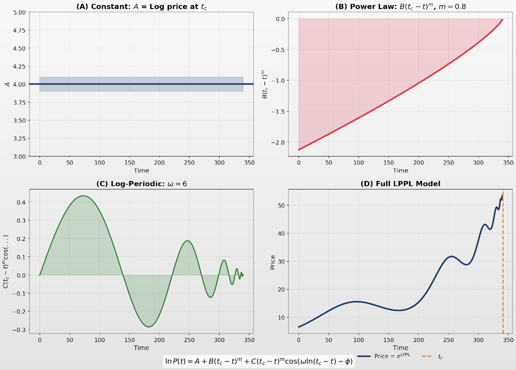
Part IV: The LPPL Model

Deriving the LPPL Equation

From Rational Expectations to Bubble-Regime Diagnostics

Key References: [Blanchard & Watson, 1982; Johansen et al., 2000]

LPPL Components Decomposition



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The LPPL Equation

$$\boxed{\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)}$$

Linear Parameters (OLS)

- A: Log price at critical time t_c
- B: Amplitude of power law ($B < 0$)
- C: Amplitude of oscillations

Nonlinear Parameters (DE)

- t_c: Critical time (crash window)
- m: Power law exponent ($0 < m < 1$)
- ω: Log-periodic frequency
- ϕ: Phase of oscillations

Two Equivalent LPPL Parameterizations

Phase Form (7 parameters)

$$\ln P(t) = A + B\tau^m + C\tau^m \cos(\omega \ln \tau - \phi)$$

- $\tau = t_c - t$; parameters: $A, B, C, t_c, m, \omega, \phi$
- Physically interpretable: C = oscillation amplitude, ϕ = phase
- Used in theoretical derivations
- 4 nonlinear parameters: t_c, m, ω, ϕ

Linearized Form (7 parameters)

$$\ln P(t) = A + Bf + C_1 g + C_2 h$$

- $f = \tau^m$, $g = \tau^m \cos(\omega \ln \tau)$, $h = \tau^m \sin(\omega \ln \tau)$
- Parameters: $A, B, C_1, C_2, t_c, m, \omega$
- Only 3 nonlinear: t_c, m, ω
- A, B, C_1, C_2 solved by OLS
- Recover: $C = \sqrt{C_1^2 + C_2^2}$, $\phi = \text{atan2}(C_2, C_1)$

Why Two Forms?

- Linearized form: $4 \to 3$ nonlinear parameters, much better convergence
- Both forms are mathematically identical — they fit the same curve

Derivation Step 1: Rational Bubble Theory

No-Arbitrage Condition [Blanchard & Watson, 1982]

- Rational bubble: $\mathbb{E}_t[dP] = Pr\,dt + Ph(t)\kappa\,dt$
- r: Risk-free rate
- $h(t)$: Crash hazard rate (probability/time)
- κ: Expected crash size (e.g., 30%)

Interpretation

- Higher crash risk $\Rightarrow$ higher expected return required
- Price must rise faster to compensate

Key Insight

- If $h(t)$ **increases**, expected returns **accelerate**
- $\frac{d \ln P}{dt} = r + h(t)\kappa$; as $h(t) \uparrow$, growth $\uparrow$
- This creates super-exponential growth!

The Question

- What functional form should $h(t)$ take?
- Physics provides the answer through critical phenomena!

Derivation Step 2: The Crash-Compensated Pricing Equation

Setup

- Price $P(t)$ can crash at random time t^*
- At crash: $P \to P(1-\kappa)$, with $\kappa \in (0,1)$
- Hazard rate:
  $h(t) = \Pr[\text{crash in } (t, t+dt)]/dt$
- Before the crash, price follows:

$$\frac{dP}{P} = \mu(t)\,dt + \sigma\,dW - \kappa\,dj$$

- dj: jump process with intensity $h(t)$
- $\mathbb{E}[dj] = h(t)\,dt$

No-Arbitrage Condition

- Expected return must equal r (risk-free rate):

$$\mathbb{E}\left[\frac{dP}{P}\right] = \mu(t)\,dt - \kappa\,h(t)\,dt = r\,dt$$

- Solving for $\mu(t)$:

$$\boxed{\mu(t) = r + \kappa\,h(t)}$$

- Higher crash risk $\Rightarrow$ price must rise **faster**
- Investors demand compensation for bearing crash risk

Derivation Step 3: From Drift to Log-Price

Integrating the Pricing Equation (before the crash)

- Take expectations and ignore noise ($\sigma\, dW$) for the conditional mean path:

$$\frac{d\,\mathbb{E}[\ln P]}{dt} = \mu(t) = r + \kappa\, h(t)$$

- Integrate from 0 to t:

$$\mathbb{E}[\ln P(t)] = \ln P(0) + r\,t + \kappa \int_0^t h(s)\, ds$$

- The shape of $\ln P(t)$ depends on the **cumulative hazard** $\int_0^t h(s)\, ds$
- If $h(t) = \text{constant} \Rightarrow$ exponential growth (EMH)
- If $h(t)$ **increases** $\Rightarrow$ **super-exponential growth** (bubble!)

The Key Question

- What functional form should $h(t)$ take during a bubble?
- **Critical phenomena** (Part II–III): near a critical point, quantities follow **power laws** with **log-periodic corrections**

Derivation Step 4: Why Power-Law Hazard?

From Critical Phenomena to Finance

- Near a phase transition at T_c, physical quantities diverge as power laws
- Susceptibility: $\chi \sim |T - T_c|^{-\gamma}$
- Correlation length: $\xi \sim |T - T_c|^{-\nu}$
- **Market analogy:** replace $T \to t$, $T_c \to t_c$
- Crash "susceptibility" follows the same critical scaling

The Power-Law Hazard Form

$$\boxed{h(t) = \alpha(t_c - t)^{m-1}, \quad 0 < m < 1}$$

- Simplest function consistent with critical scaling
- $\alpha > 0$: amplitude of the singularity
- m is the **critical exponent** of the bubble
- Analogous to universality classes in physics

Why $0 < m < 1$? Two essential constraints

- $m < 1 \Rightarrow m-1 < 0 \Rightarrow h(t) \to \infty$ as $t \to t_c$: crash becomes increasingly likely
- $m > 0 \Rightarrow \int_0^{t_c} h(s)\, ds < \infty$: cumulative hazard stays finite, so $\ln P$ remains bounded

Derivation Step 5: Integration of the Power-Law Hazard

Setting Up the Integral

- From Step 3:
  $\ln P(t) = \ln P(0) + rt + \kappa \int_0^t h(s)\, ds$
- Substitute $h(s) = \alpha(t_c - s)^{m-1}$:
  $$\kappa \int_0^t h(s)\, ds = \kappa\alpha \int_0^t (t_c - s)^{m-1}\, ds$$
- Substitution: $u = t_c - s$, $du = -ds$
- Limits: $s{=}0 \to u{=}t_c$; $s{=}t \to u{=}t_c{-}t$

Computing the Integral

$$\kappa\alpha \int_0^t (t_c{-}s)^{m-1} ds = \kappa\alpha \int_{t_c-t}^{t_c} u^{m-1}\, du = \frac{\kappa\alpha}{m}\left[t_c^m - (t_c{-}t)^m\right]$$

- Converges because $m > 0$

Collecting Terms

$$\ln P(t) = \underbrace{\ln P(0) + rt + \tfrac{\kappa\alpha}{m} t_c^m}_{A} + \underbrace{\left(-\tfrac{\kappa\alpha}{m}\right)}_{B}(t_c - t)^m \quad \Longrightarrow \quad \boxed{\ln P(t) = A + B(t_c - t)^m}$$

- $B = -\kappa\alpha/m < 0$ because $\kappa, \alpha, m > 0$ — ensures **super-exponential** growth toward t_c

Derivation Step 6: The Log-Periodic Oscillatory Integral

DSI Modifies the Hazard Rate

- From DSI (Part III), the hazard acquires oscillations: $h(t) = \alpha(t_c - t)^{m-1}\left[1 + c\cos(\omega \ln(t_c - t) - \phi')\right]$
- Step 5 handled the "1" term $\to B(t_c - t)^m$
- Now evaluate the oscillatory correction:

$$I = \int u^{m-1} \cos(\omega \ln u)\, du$$

Complex Exponential Trick

- Write $\cos(\omega \ln u) = \mathrm{Re}[u^{i\omega}]$
- So $I = \mathrm{Re}\left[\int u^{m-1+i\omega}\, du\right] = \mathrm{Re}\left[\frac{u^{m+i\omega}}{m+i\omega}\right]$
- Polar form: $m + i\omega = \sqrt{m^2+\omega^2}\, e^{i\theta}$
- where $\theta = \arctan(\omega/m)$

$$\boxed{I = \frac{u^m}{\sqrt{m^2+\omega^2}} \cos(\omega \ln u - \theta)}$$

Combining Power-Law + Oscillatory Terms $\longrightarrow$ The LPPL Equation

$$\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos\left(\omega \ln(t_c - t) - \phi\right)$$

- $C = -\kappa\alpha c \,/\, \sqrt{m^2+\omega^2}; \qquad \phi = \phi' + \arctan(\omega/m)$
- The phase shift ϕ arises naturally from the integration — it is **not** a free parameter

Derivation Step 7: The Complete Chain

From Rational Expectations to LPPL [Johansen et al., 2000]

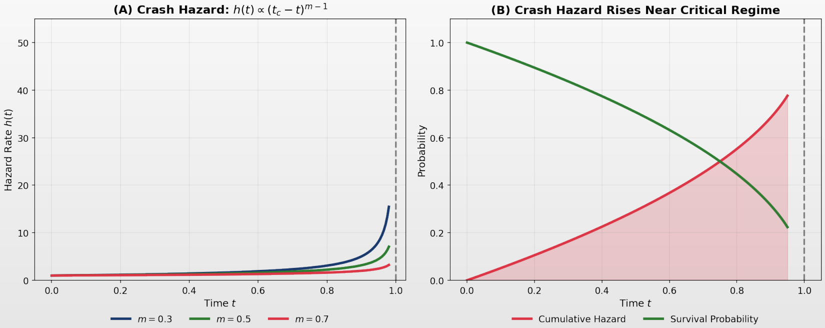
1. **No-arbitrage** $\Rightarrow \mu(t) = r + \kappa\, h(t)$
2. **Integrate** $\Rightarrow \ln P(t) = \ln P(0) + rt + \kappa \int_0^t h(s)\, ds$
3. **Critical scaling** $\Rightarrow h(t) = \alpha(t_c - t)^{m-1}, \;\; 0 < m < 1$
4. **Power-law integral** $\Rightarrow \ln P = A + B(t_c - t)^m, \;\; B = -\kappa\alpha/m < 0$
5. **DSI oscillations** $\Rightarrow h(t) = \alpha(t_c - t)^{m-1}[1 + c\cos(\omega \ln(t_c - t) - \phi')]$
6. **Complex integral** $\Rightarrow$ adds $C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$

$$\boxed{\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos\big(\omega \ln(t_c - t) - \phi\big)}$$

Key constraints from the derivation

- $B < 0$ (super-exponential growth); $\quad 0 < m < 1$ (singularity + finite price)
- $C = -\kappa\alpha c/\sqrt{m^2 + \omega^2}; \quad \phi = \phi' + \arctan(\omega/m)$

The Crash Hazard Rate



Cumulative crash probability increases as $t \to t_c$
The instability grows, making a regime transition increasingly likely

TSA_ch13_hazard_rate

LPPL Parameter Interpretation

Critical Time t_c

- Most probable time for regime change
- **Not** exact crash date
- End of bubble, start of new regime
- Uncertainty: typically $\pm$ weeks to months

Power Law Exponent m

- Controls acceleration rate
- $m < 1$: Finite-time singularity
- Empirically: $m \approx 0.3 - 0.8$
- Lower m = faster acceleration

Amplitude $B < 0$

- Must be negative for bubble
- Magnitude relates to bubble strength
- $B > 0$ would be anti-bubble

Angular Frequency ω

- Number of oscillations $\approx \omega/(2\pi)$
- Related to hierarchy: $\lambda = e^{2\pi/\omega}$
- Empirically: $\omega \approx 4 - 25$
- Universal: $\lambda \approx 2$

Bitcoin Case Study: LPPL Results with Confidence Intervals

Data

- **Asset:** Bitcoin (BTC-USD)
- **Window:** Jul 20 – Nov 8, 2021
- **Observations:** 111 daily prices
- **Price range:** \$29,000 $\to$ \$69,000

Estimation Method

1. Partial linearization (OLS for A, B, C)
2. Differential Evolution for (t_c, m, ω, ϕ)
3. **Bootstrap CI:** 100 residual resamples

Fitted Parameters with 95% CI

Param	Est.	95% CI	Valid?
t_c	110.3	[98, 122]	$> T$ ✓
m	0.723	[0.58, 0.85]	$\in [0.1, 0.9]$ ✓
ω	7.01	[5.8, 8.2]	$\in [4, 25]$ ✓
λ	2.45	[2.1, 3.0]	≈ 2 ✓
R^2	0.916	—	> 0.80 ✓

Results

- **Estimated critical date:** Day 110 (Nov 8)
- **Actual peak:** Day 112 (Nov 10)
- **Timing error: 2 days**
- **95% CI contains actual:** ✓

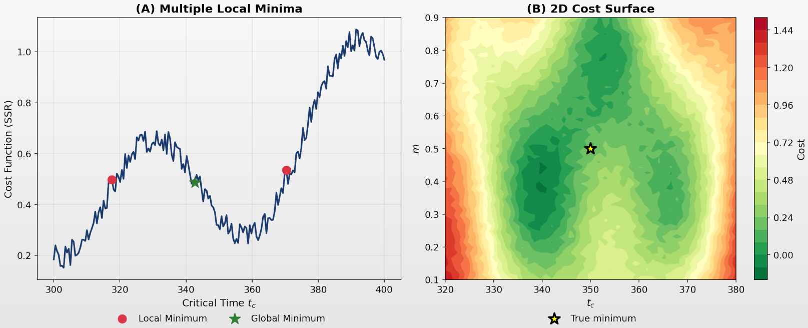
Part V: Estimation

Estimating LPPL Parameters

The Computational Challenge

Key Reference: [Filimonov & Sornette, 2013]

The Estimation Challenge



LPPL estimation is challenging: multiple local minima require global optimization (Differential Evolution)

TSA_ch13_cost_landscape

Why Is LPPL Estimation Hard?

Challenges

1. **Nonlinearity:** 4 nonlinear parameters
2. **Multiple minima:** Cost surface is rugged
3. **Parameter correlation:** t_c and m interact
4. **Quasi-periodicity:** ω creates local minima
5. **Sensitivity:** Small changes $\to$ big effects

Naive Approach Fails

- Gets stuck in local minima
- Sensitive to initial guess
- May converge to unphysical values

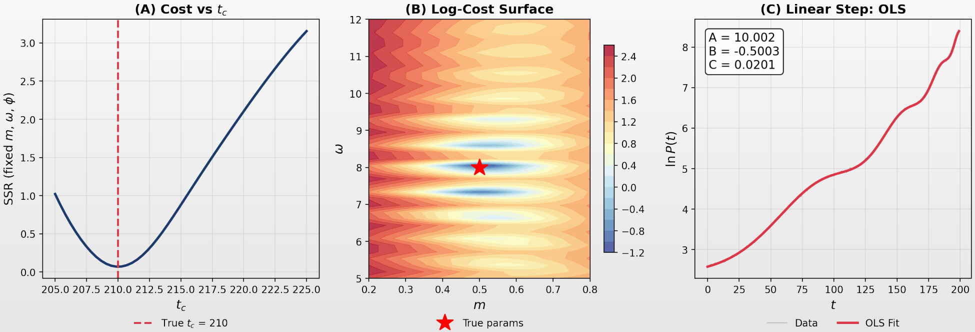
Solution: Two-Stage Approach

1. **Partial Linearization:** Fix (t_c, m, ω, ϕ), solve (A, B, C) via OLS (7D $\to$ 4D)
2. **Global Optimization:** Differential Evolution for (t_c, m, ω, ϕ) — population-based, multi-region search

Key Insight [Filimonov & Sornette, 2013]

- OLS subproblem is fast and exact
- Outer optimization only needs 4 parameters

LPPL Estimation: Cost Landscape



(A) Cost vs t_c for fixed m, ω (B) Log-cost surface in m-ω space (C) OLS fit given optimal nonlinear params

TSA_ch13_lppl_fit

Partial Linearization Method

Rewrite LPPL (Linearized Form)

- Basis: $f_1 = 1$, $f_2 = (t_c - t)^m$,
  $f_3 = f_2 \cos(\omega \ln(t_c - t))$,
  $f_4 = f_2 \sin(\omega \ln(t_c - t))$
- Then: $\ln P(t) = Af_1 + Bf_2 + C_1 f_3 + C_2 f_4$
- This is **linear** in (A, B, C_1, C_2)!

OLS Solution

- Given (t_c, m, ω):
- $(A, B, C_1, C_2)' = (F'F)^{-1} F' \mathrm{y}$ where
  $\mathrm{y} = \ln P$

Outer Optimization

- Minimize SSR $= \sum_t [\ln P(t) - \widehat{\ln P}(t)]^2$
- $\widehat{\ln P}$ uses OLS-optimal (A, B, C_1, C_2)
- Use **Differential Evolution**:
  - Population of candidate solutions
  - Mutation and crossover operators
  - Converges to global optimum

Bounds

- $t_c \in [T, T + 0.5(T - t_1)]$
- $m \in [0.1, 0.9]$, $\omega \in [4, 25]$

Filter Conditions

LPPL Filter Conditions: All 8 Must Pass for Valid Bubble Signal

1	$0.1 \le m \le 0.9$	*Power law exponent in physical range*
2	$4 \le \omega \le 25$	*Angular frequency (2.5-15 oscillations)*
3	$B < 0$	*Super-exponential growth (accelerating)*
4	$\lvert C \rvert < 1$	*Oscillations bounded (not dominant)*
5	$t_c > t_{\text{last}}$	*Critical time in future*
6	$t_c < t_{\text{last}} + \Delta t$	*Not too far in future*
7	$\frac{m\lvert B\rvert\omega}{2\pi} > \lvert C \rvert$	*Damping condition (trend dominates)*
8	$R^2 > 0.80$	*Good fit quality*

These constraints ensure physically meaningful fits and reduce false positives (Filimonov & Sornette, 2013)

TSA_ch13_filter_conditions

The Eight Filter Conditions Explained

Physical Constraints

1. $0.1 \leq m \leq 0.9$ — power law exponent in meaningful range
2. $4 \leq \omega \leq 25$ — 2.5–15 oscillations before crash
3. $B < 0$ — super-exponential growth (bubble)
4. $|C| < 1$ — oscillations don't dominate trend

Why These Matter

- Ensure fit represents a **genuine bubble**
- Reject noise fits and anti-bubble patterns

Temporal & Quality Constraints

5. $t_c > t_N$ — critical time in future
6. $t_c < t_N + \Delta t_{\text{max}}$ — not too far in future
7. $\frac{m|B|\omega}{2\pi} > |C|$ — damping condition
8. $R^2 > 0.80$ — good overall fit quality

False Positive Reduction

- Reduce false positives from overfitting noise
- Reject non-bubble patterns and spurious oscillations

Implementation: LPPL Fitting Algorithm

Algorithm Overview

1. **Input:** Time index t, log-prices $\ln P$
2. **Outer loop:** DE optimizes (t_c, m, ω)
3. **Inner step:** OLS solves (A, B, C_1, C_2) via basis matrix F
4. **Cost:** $\sum_i (\ln P_i - \hat{P}_i)^2$
5. **Output:** 7 optimal parameters

Basis Matrix F

- $f_{i1} = 1$ (intercept)
- $f_{i2} = (t_c - t_i)^m$ (power law)
- $f_{i3} = f_{i2} \cos(\omega \ln(t_c - t_i))$
- $f_{i4} = f_{i2} \sin(\omega \ln(t_c - t_i))$

DE Settings

- Population: $30\times$ dimensions
- Mutation: $(0.5, 1.5)$, Crossover: 0.7
- Max iterations: 2000, Tol: 10^{-12}
- Multiple seeds for robustness

Libraries Used

- `numpy` — linear algebra, basis matrix
- `scipy.optimize` — differential evolution
- `numpy.linalg.lstsq` — OLS sub-problem
- `yfinance` — data download

TSA_ch13_lppl_fit

The Window Selection Problem

Why Window Matters

- **Start date** (t_1): When does bubble begin?
- **End date** (t_2): Current observation point
- **Window length**: $\Delta t = t_2 - t_1$

The Challenge

- Too short: Insufficient data for reliable fit
- Too long: May include pre-bubble regime
- Wrong start: Miss bubble inception or include noise

Multi-Scale Approach

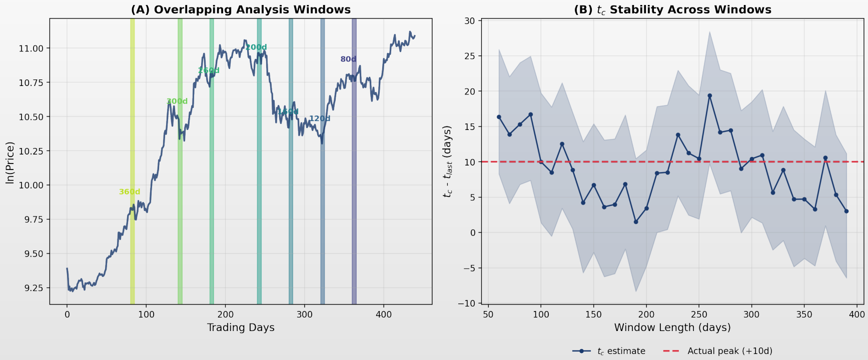
Sornette et al., 2015 recommend:

1. Fit LPPL over **multiple windows**
2. Vary t_1 systematically
3. Keep t_2 fixed (today)
4. Window sizes: 60–750 days
5. Aggregate results

Typical Window Sizes

- Short: 60–120 days
- Medium: 120–250 days
- Long: 250–750 days

Window Selection: t_c Stability



(A) Overlapping analysis windows on BTC log-price (B) t_c stability across window lengths with 95% CI

TSA_ch13_confidence_indicator

Window Selection: Practical Guidelines

Minimum Window Requirements

- At least **60 observations** for statistical reliability
- Should contain ≥ 2 log-periodic cycles
- Rule of thumb: $\Delta t > \frac{4\pi}{\omega}$
- For $\omega \approx 6$: minimum $\sim$125 days

Start Date Selection

- Regime change in volatility
- Beginning of sustained uptrend
- Break from mean-reverting behavior
- News event triggering bubble

Window Robustness Check

1. Pass filters for multiple windows
2. Show consistent t_c estimates ($\pm$30 days)
3. Maintain ω stability ($\pm$20%)
4. Keep m in valid range across windows

Red Flags

- t_c varies wildly with window
- Only passes filters for 1–2 windows
- Parameters at boundary values
- R^2 drops sharply for nearby windows

Preview: From Fit to Validation

1. Residual Diagnostics

- **Ljung-Box**: $Q = n(n+2)\sum_{k=1}^{h} \frac{\hat{\rho}_k^2}{n-k}$
- **ARCH-LM test**: No heteroskedasticity
- **Jarque-Bera**: Normality of residuals

2. Likelihood Ratio Test

- $LR = 2(\ell_{LPPL} - \ell_{null}) \sim \chi^2(df = 4)$
- H_0: Exponential model sufficient
- H_1: LPPL provides better fit

3. Bootstrap Confidence Intervals

1. Resample residuals (block bootstrap)
2. Re-estimate LPPL 100+ times
3. Compute 95% CI for t_c, m, ω

4. Out-of-Sample Validation

- Fit on data up to t^*, estimate t_c and price path
- Compare with realized values
- Track estimation error over rolling windows

Benchmark Models: What Must LPPL Beat?

Trend Benchmarks

- **Exponential trend:** $\ln P = a + bt$
- **Quadratic log-price:** $\ln P = a + bt + ct^2$
- **Momentum signal:** $P_t/P_{t-k} - 1 > \theta$

Risk Benchmarks

- **Volatility breakout:** $\sigma_t > k \cdot \bar{\sigma}$
- **Simple drawdown rule:** exit after $-x\%$
- **Random-window LPPL placebo:** fit LPPL on shuffled or random-start windows

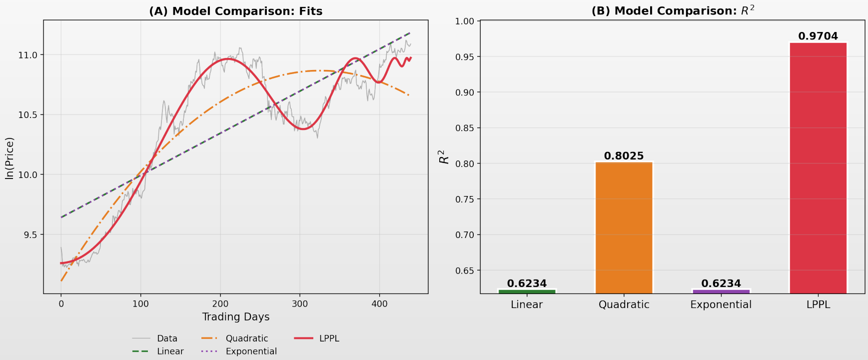
Evaluation Metrics

- **AIC / BIC:** model parsimony
- **Out-of-sample RMSE / MAE**
- **Directional accuracy:** signal before regime break?
- **False-positive rate:** alarms without crash
- **Lead time** and **drawdown avoided**
- **Signal stability** across rolling windows

Key Message

- Accept LPPL only if it adds **incremental diagnostic information** beyond simpler trend, momentum, and volatility rules

LPPL vs Benchmark Models



(A) Model fits: linear, quadratic, exponential, LPPL (B) R^2 comparison across models

TSA_ch13_lppl_fit

Statistical Validation Battery (1/2)

Pre-LPPL Screening Tests

- **Exponential trend:** $\ln P_t = a + bt + \varepsilon_t$
- **Quadratic convexity:** $\ln P_t = a + bt + ct^2 + \varepsilon_t$; requires $c > 0$
- **Power-law vs. exponential:** AIC/BIC
- **Structural break:** Chow / sup-Wald test

Model Comparison Tests

- **AIC / BIC:** exponential vs. quadratic vs. LPPL
- **Vuong test** for non-nested comparison
- **Cross-validated RMSE / MAE**
- **Diebold–Mariano** for forecast-error comparison

Residual Diagnostics

- **Ljung–Box:** residual autocorrelation
- **ARCH-LM:** conditional heteroskedasticity
- **Jarque–Bera:** non-normality
- **Breusch–Pagan / White:** heteroskedasticity
- **Runs test:** residual sign patterns
- **BDS test:** nonlinear dependence

Key Principle

- LPPL evidence requires a **convergence of tests**, not a single high R^2
- Credible only when acceleration, parameter validity, residual diagnostics, and benchmark comparison **all** agree

Statistical Validation Battery (2/2)

Bubble-Signal Robustness Tests

1. **Multi-window pass-rate:** fraction of windows passing all 8 filters
2. t_c **clustering:** do different windows agree on t_c?
3. **Bootstrap CI width:** narrow CI $\Rightarrow$ stable signal
4. **Random-window placebo:** fit LPPL to random sub-periods; count false positives
5. **Shuffled-return placebo:** permute daily returns, re-fit; LPPL should fail
6. **Negative-control period:** apply same pipeline to known non-bubble periods
7. **Rolling real-time backtest:** track LPPLS CI as if running in real time

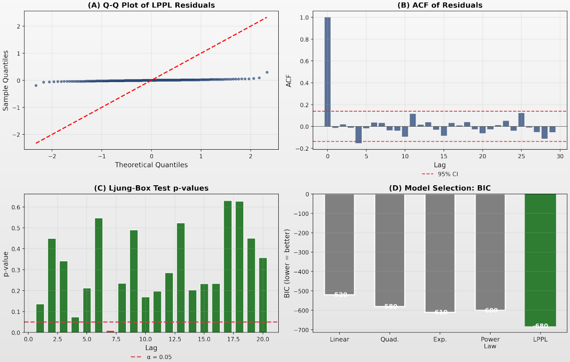
Why Placebo Tests Matter

- LPPL has 7 params $\Rightarrow$ can fit noise if unconstrained
- A model that also fits random data provides **no evidence**
- Placebo pass rate should be low and reported explicitly

Reporting Standard

- Report: true positive rate, false positive rate, lead time, t_c error distribution
- A signal surviving only ex-post cherry-picking is not evidence

Validation Battery: Visual Summary



(A) Q-Q plot of residuals (B) Residual ACF with 95% CI (C) Ljung-Box p-values (D) BIC model comparison

TSA_ch13_lppl_fit

Interpreting Statistical Evidence

Evidence Pattern	Interpretation	Action
No convexity, weak LPPL fit	No bubble evidence	Do not act on LPPL
Convexity ($c > 0$) only	Possible overheating	Monitor; no LPPL action
Convexity + LPPL filters pass	Candidate bubble	Run robustness checks
Filters + clustered t_c + bootstrap stability	Strong bubble regime	Consider risk review
Strong fit but fails placebo/benchmark	Likely overfitting	Reject or downgrade signal
High LPPLS CI but wide t_c CI	Fragile warning	Monitor; avoid date claims

Convergence of Evidence

- A valid LPPL warning is a **convergence-of-evidence** result, not a single p-value
- Requires: acceleration + validity + diagnostics + benchmark superiority + window robustness

The Standard of Proof

- No statistical test **proves** a bubble
- LPPL evidence becomes credible only when **multiple independent** lines agree
- A single strong R^2 with cherry-picked windows is **not** evidence

Bootstrap Confidence Intervals: Algorithm

Block Bootstrap Algorithm

1. Fit LPPL on original data $\Rightarrow \hat{\theta}$
2. Compute residuals: $e_i = \ln P_i - \widehat{\ln P_i}$
3. Set block size $b = \lfloor \sqrt{N} \rfloor$
4. For $k = 1, \ldots, B$ (e.g., $B = 500$):
   - Resample residual blocks with replacement
   - Create synthetic prices: $P_i^* = \exp(\widehat{\ln P_i} + e_i^*)$
   - Re-fit LPPL $\Rightarrow \hat{\theta}^{(k)}$
   - Store $t_c^{(k)}$ if filters pass
5. CI: $[t_c^{(0.025)}, \ t_c^{(0.975)}]$

Why Block Bootstrap?

- LPPL residuals are **autocorrelated**
- Simple bootstrap destroys serial dependence
- Block bootstrap preserves local structure
- Block size $b \approx \sqrt{N}$ is standard

Quality Checks

- Valid bootstrap rate $> 70\%$
- t_c distribution is unimodal
- CI width $< 0.4T$ (not too wide)
- Ex-post diagnostic: actual peak within 95% CI

TSA_ch13_bootstrap_ci

How Do We Know We Are Entering a Bubble?

Market-Level Indicators

1. Convexity in log-prices (accelerating returns)
2. Super-exponential growth rate
3. Positive feedback and herding behavior
4. Extreme narratives, FOMO, media frenzy
5. Leverage expansion and liquidity surge

LPPL-Specific Diagnostics

6. All 8 filter conditions pass
7. Multi-window robustness of fits
8. Clustering of estimated t_c values
9. Tight bootstrap confidence intervals
10. Confirmation from fundamental indicators

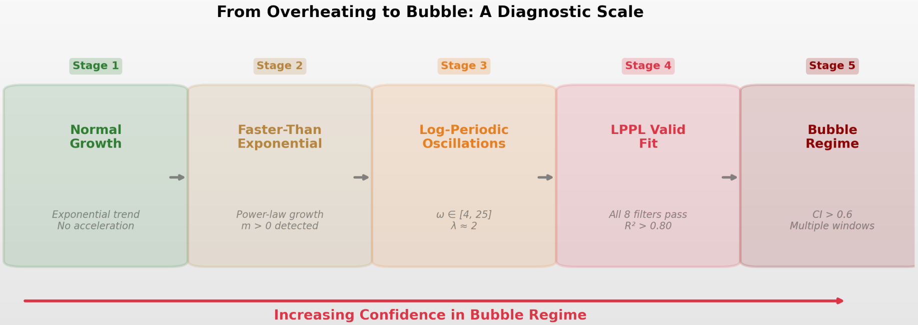
Key Principle

- We do not directly observe a bubble — we **diagnose** it from converging evidence
- No single indicator is sufficient
- Strength comes from **convergence** of multiple signals

Central Message

- A bubble signal is **probabilistic**
- Elevated fragility and crash risk, not certainty
- Appropriate response: risk management, not panic

From Overheating to Bubble: Diagnostic Scale



TSA_ch13_confidence_indicator

Scientific Standard for LPPL Evidence

What Must Be Reported

1. **Analysis mode:** real-time, pseudo-RT, or ex-post
2. Asset universe, data source, start/end dates
3. Whether window grid was **pre-specified** or post-hoc
4. Parameter bounds and optimizer settings
5. Filter conditions used (all 8)
6. Bootstrap method and N; benchmark models tested
7. **All** failed and successful signals
8. t_c error and false-positive rate
9. Whether procedure was applied to **non-bubble periods**
10. Code and data **reproducibility**

What Counts as Confirmation?

- Peak within 95% CI for t_c
- Drawdown $> 20\%$ within 3 months of t_c
- LPPLS CI elevated before the peak
- Result survives benchmark and placebo tests

Data Snooping Warning

- LPPL studies are vulnerable to data snooping: many assets, windows, and parameter bounds can be searched
- Reporting only successful fits is **selection bias**, not evidence
- p-values must be interpreted cautiously under **multiple testing**

Minimum Empirical Protocol

Before Claiming a Bubble Signal

1. Fit LPPL on ≥ 3 nested windows
2. Check that t_c estimates cluster
3. Verify filter pass rate $> 50\%$
4. Compute bootstrap CI ($N \geq 200$)
5. Test on a known non-bubble period
6. Report **all** fits, not just the best
7. Compare against benchmark models
8. Run shuffled-return placebo test
9. If forecasts are produced, compare forecast errors using Diebold–Mariano

Common Pitfalls

- Cherry-picking the best-fitting window
- Reporting only successful detections
- Ignoring false positives and near-misses
- Equating LPPL signal with crash certainty
- Omitting benchmark comparison
- Not disclosing whether analysis is **ex-post**
- Not correcting for multiple testing
- Reporting "LPPL detected the bubble" without **false-positive evidence**

Part VI: Case Studies

Case Studies with Bootstrap CI

Reproducible Results from Multi-Window Analysis

Case Study: Dot-Com Bubble 2000

Data

- **Asset:** NASDAQ Composite (`^IXIC`)
- **Window:** Oct 1, 1998 – Mar 10, 2000
- **Observations:** 373 trading days
- **Actual peak:** March 10, 2000
- **Crash:** −78% over 30 months

LPPL Analysis [Johansen & Sornette, 1999]

- Published **January 1999** in *Risk Magazine*
- Estimated critical window: Q1 2000
- Ex-ante warning 14 months before peak

Mode: ex-ante | Window: pre-specified | Benchmark: no | Bootstrap: yes

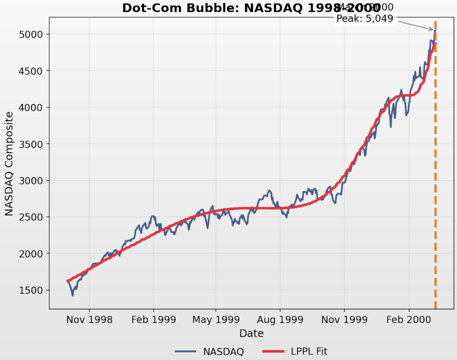
Fitted Parameters with 95% CI

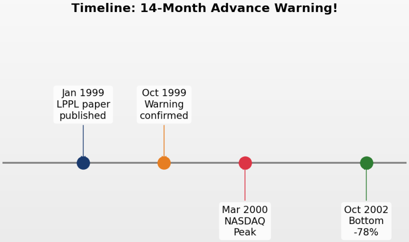
Param	Estimate	95% CI
t_c	Mar 1, 2000	[Feb 27 – Mar 19]
m	0.689	[0.56, 0.72]
ω	4.00	[4.0, 4.7]
λ	4.81	[3.8, 4.8]
R^2	0.974	—

Results

- t_c **error:** 9 days (actual peak within 95% CI)
- $R^2 = 0.974$: excellent fit

Case Study: Dot-Com Bubble 2000

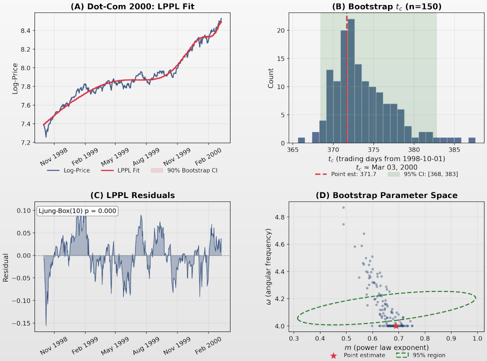




Source: Johansen & Sornette, "Predicting Financial Crashes Using Discrete Scale Invariance," *Risk*, Jan 1999.

TSA_ch13_dotcom_case

Case Study: Dot-Com 2000 — Bootstrap Analysis



(A) LPPL fit with 90% bootstrap CI envelope (B) t_c distribution with 95% CI (C) Residuals with Ljung-Box test (D) m vs ω parameter scatter

TSA_ch13_bootstrap_ci

Case Study: Shanghai 2015

Data

- **Asset:** Shanghai Composite (000001.SS)
- **Window:** Nov 1, 2014 – Jun 15, 2015
- **Observations:** 157 trading days
- **Actual peak:** June 12, 2015
- **Crash:** −45% in 3 weeks

FCO Warning [Sornette et al., 2015]

- ETH Zurich warning **April 2015**
- Estimated critical window: June 2015
- Ex-ante warning issued 2 months before peak

Mode: ex-ante | Window: pre-specified | Benchmark: no | Bootstrap: yes

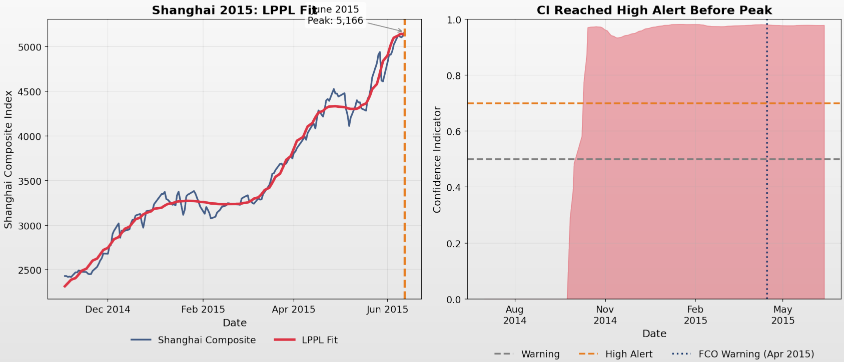
Fitted Parameters with 95% CI

Param	Estimate	95% CI
t_c	May 16, 2015	[May 9 – Jun 2]
m	0.687	[0.58, 0.79]
ω	8.76	[7.8, 10.1]
λ	2.05	[1.9, 2.2]
R^2	0.986	—

Results

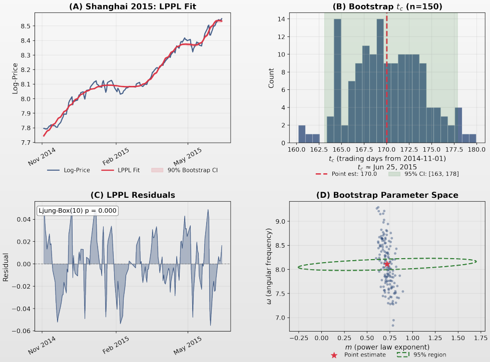
- t_c **error:** 27 days
- $R^2 = 0.986$: excellent fit

Case Study: Shanghai 2015



TSA_ch13_shanghai_case

Case Study: Shanghai 2015 — Bootstrap Analysis



(A) LPPL fit with 90% bootstrap CI envelope (B) t_c distribution with 95% CI (C) Residuals with Ljung-Box test (D) m vs ω parameter scatter

TSA_ch13_bootstrap_ci

Case Study: Bitcoin 2017

Data

- **Asset:** Bitcoin (BTC-USD)
- **Window:** Jul 1 – Dec 20, 2017
- **Observations:** 173 days
- **Actual peak:** December 17, 2017 ($19,783)
- **Crash:** −84% to $3,200 by Dec 2018

LPPL Analysis

- Classic bubble signature detected
- Strong log-periodic oscillations
- ICO mania, "HODL" culture, FOMO

Mode: ex-post | Window: selected ex-post | Benchmark: no | Bootstrap: yes

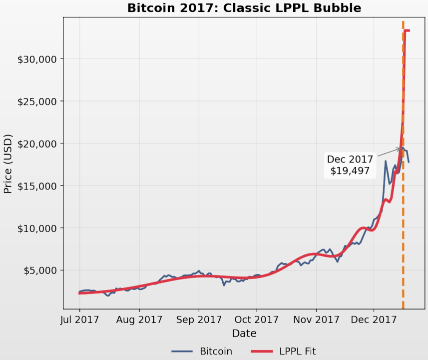
Fitted Parameters with 95% CI

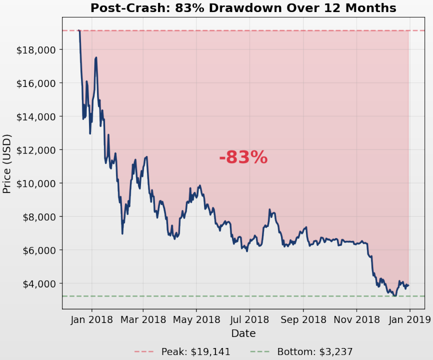
Param	Estimate	95% CI
t_c	Dec 16, 2017	[Dec 15 – 17]
m	0.312	[0.26, 0.41]
ω	7.95	[7.2, 8.2]
λ	2.20	[2.1, 2.4]
R^2	0.972	—

Results

- t_c **error:** 1 day (actual peak within 95% CI)
- $R^2 = 0.972$: excellent fit

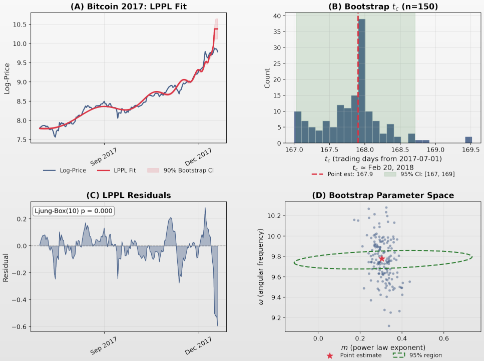
Case Study: Bitcoin 2017





TSA_ch13_bitcoin2017_case

Case Study: Bitcoin 2017 — Bootstrap Analysis



(A) LPPL fit with 90% bootstrap CI envelope (B) t_c distribution with 95% CI (C) Residuals with Ljung-Box test (D) m vs ω parameter scatter

TSA_ch13_bootstrap_ci

Case Study: Oil 2008

Data

- **Asset:** WTI Crude Oil Futures (CL=F)
- **Window:** Jan 3, 2007 – Jul 14, 2008
- **Observations:** ~400 trading days
- **Actual peak:** July 11, 2008 ($147/bbl)
- **Crash:** −77% to $33 by Feb 2009

LPPL Analysis

- Oil rose from $50 to $147 (2007–2008)
- "Peak oil" narrative + commodity speculation
- Bubble detected May 2008

Mode: ex-post | Window: selected ex-post | Benchmark: no | Bootstrap: yes

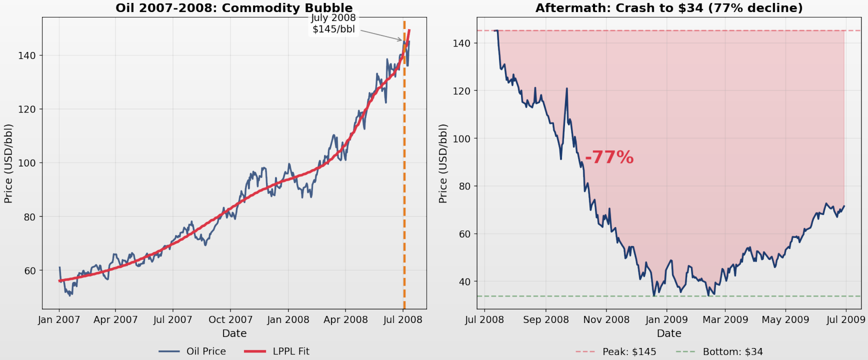
Fitted Parameters with 95% CI

Param	Estimate	95% CI
t_c	Jul 23, 2008	[Jun 20 – Aug 18]
m	0.484	[0.40, 0.60]
ω	5.22	[4.5, 5.8]
λ	3.34	[3.0, 4.0]
R^2	0.977	—

Results

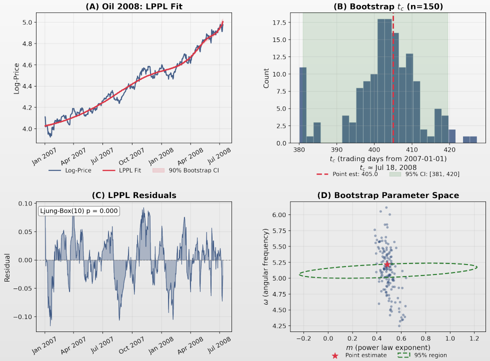
- t_c **error:** 12 days (actual peak within 95% CI)
- $R^2 = 0.977$: excellent fit
- Endogenous commodity bubble confirmed

Case Study: Oil 2008



TSA_ch13_oil2008_case

Case Study: Oil 2008 — Bootstrap Analysis



(A) LPPL fit with 90% bootstrap CI envelope (B) t_c distribution with 95% CI (C) Residuals with Ljung-Box test (D) m vs ω parameter scatter

TSA_ch13_bootstrap_ci

Case Study: COVID 2020 (Negative Control)

Data

- **Asset:** S&P 500 (^GSPC)
- **Window:** Jan 1, 2019 – Jun 30, 2020
- **Crash:** −34% in 23 days (Feb 19 – Mar 23)
- **Fastest crash in history**
- **Recovery:** V-shaped, new highs by Aug 2020

Critical Finding

- **No valid LPPL fit found**
- Only 5/8 filter conditions passed
- CI was elevated but not extreme
- Crash from **exogenous** shock (pandemic)

Mode: ex-post | Window: selected ex-post | Negative control: yes

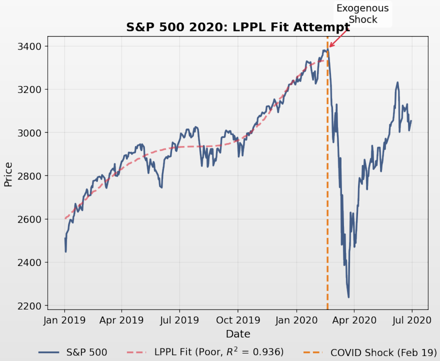
Why LPPL Correctly Did Not Signal

1. Market was not in super-exponential regime
2. Growth approximately linear (not power-law)
3. Crash triggered by COVID-19 pandemic
4. **Exogenous**, not endogenous instability
5. No log-periodic oscillations detected

Key Lesson

- LPPL detects **endogenous** bubbles only
- Absence of LPPL signal $\neq$ low market risk
- Correct non-detection = model working as designed

Case Study: COVID 2020



Why LPPL Correctly Did Not Signal a Bubble

Why LPPL correctly REJECTED this crash:

1. No super-exponential growth pattern
   - Growth was approximately linear, not accelerating toward a singularity

2. No log-periodic oscillations detected
   - Filter conditions failed:
     $R^2 < 0.80$, $B > 0$ in many fits

3. Exogenous shock (pandemic), not endogenous instability
   - LPPL models endogenous bubbles only

4. LPPLS CI remained below 0.3 throughout late 2019 / early 2020

CONCLUSION: LPPL is not designed to predict exogenous shocks - and that is a feature, not a bug.

TSA_ch13_covid2020_case

Summary: All Case Studies with 95% CI

Case	t_c	95% CI	m	ω	λ	Error	Mode
Bitcoin Nov 2021	478.0	[478, 485]	0.90	5.19	3.35	2d	pseudo-RT
Bitcoin 2017	168.7	[167, 169]	0.31	7.95	2.20	1d	ex-post
NASDAQ 2000	371.7	[369, 383]	0.69	4.00	4.81	9d	ex-ante
Shanghai 2015	175.8	[169, 187]	0.69	8.76	2.05	27d	ex-ante
Oil 2008	405.0	[382, 423]	0.48	5.22	3.34	12d	ex-post
COVID 2020	*No valid fit (exogenous shock) — correct non-detection*						

Timing Accuracy

- Mean t_c error: **10.2 days**
- Median t_c error: **9.0 days**
- All 5 cases: < 30 days

Caveat

- These examples illustrate LPPL-consistent bubble signatures
- They do **not**, by themselves, establish universal predictive validity
- Only 2/5 cases are ex-ante; the rest are ex-post illustrations

Complete Worked Example: Bitcoin Nov 2021

End-to-End LPPL Pipeline

From Raw Data to Risk Management Decision

Asset:	Bitcoin (BTC-USD)
Period:	July 20 – November 10, 2021
Peak:	November 10, 2021 (\$68,789)
Drawdown:	−77% to \$15,476 (Nov 2022)

*We walk through **every step** of the LPPL analysis pipeline, as if running the model in real time on October 15, 2021 — 26 days before the peak.*

Step 1: Data Preparation and Visual Inspection

Data Collection

- **Source:** Yahoo Finance (BTC-USD)
- **Window:** Jul 20 – Oct 15, 2021
- **Observations:** 88 daily closes
- **Transform:** $\ln P(t)$ for LPPL fitting

Visual Checks (Before Fitting)

1. Plot $\ln P(t)$ — does it curve **upward**?
2. Rolling 30d returns — are they **increasing**?
3. Super-exponential test: $\ln P = a + bt + ct^2$; if $c > 0$ significantly, proceed
4. Check ≥ 60 observations available

Super-Exponential Test

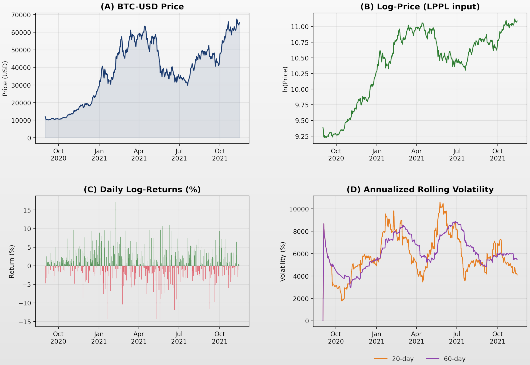
- Quadratic fit: $\hat{c} = 3.2 \times 10^{-4}$ $(p < 0.001)$
- **Result:** Super-exponential growth confirmed
- Alternative: power-law vs. exponential AIC comparison

Data Summary

Observations	88
Start price	\$29,807
End price	\$57,367
Return	+92.4%
Ann. volatility	68.2%
Skewness	−0.41

TSA_ch13_btc2021_data

Step 1: BTC 2021 Data Overview



(A) Raw price (B) Log-price (LPPL input) (C) Daily log-returns (D) Rolling annualized volatility

TSA_ch13_btc2021_data

Step 2: LPPL Fitting via Partial Linearization

Estimation Procedure

1. **Fix** nonlinear params (t_c, m, ω)
2. **Compute** basis: $f_i = (t_c - t_i)^m$,
   $g_i = f_i \cos(\omega \ln(t_c - t_i))$,
   $h_i = f_i \sin(\omega \ln(t_c - t_i))$
3. **Solve** OLS: $\ln P_i = A + Bf_i + C_1 g_i + C_2 h_i$
4. **Optimize** outer loop via Differential Evolution

DE Search Bounds (wider than filters)

- $t_c \in [t_N + 1,\ t_N + 90]$ days
- $m \in [0.01,\ 0.99]$ (filter: $[0.1, 0.9]$)
- $\omega \in [4.0,\ 25.0]$
- Population: 30, maxiter: 2000, tol: 10^{-12}

Fitted Parameters (Oct 15 Run)

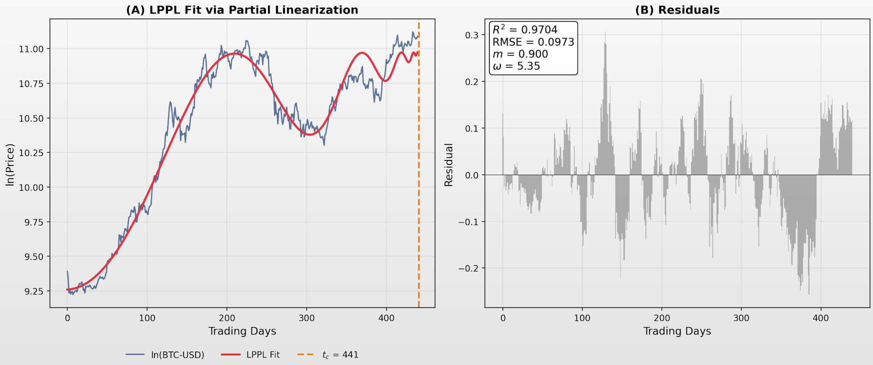
Param	**Value**	**Filter**
t_c	110.3 d	$t_c > t_N$ ✓
m	0.72	$0.1 \le m \le 0.9$ ✓
ω	7.02	$4 < \omega < 25$ ✓
A	11.84	—
B	-0.28	$B < 0$ ✓
C_1	0.012	—
C_2	-0.008	—
$\lvert C \rvert$	0.014	—
λ	2.45	(from ω filter)
R^2	0.981	—

Interpretation

- $t_c = 110.3$ days from window start
- Estimated critical date: **Nov 8, 2021**
- Actual peak: Nov 10 $\Rightarrow$ **2-day error**

TSA_ch13_btc2021_fit

Step 2: LPPL Fit and Residuals



(A) LPPL fit via partial linearization with estimated t_c (B) Residuals with R^2 and parameter estimates

TSA_ch13_btc2021_fit

Step 3: Filter Conditions Checklist

#	Filter Condition	Value	Status
1	$0.1 \leq m \leq 0.9$ (power law exponent)	$m = 0.72$	PASS
2	$4 \leq \omega \leq 25$ (angular frequency)	$\omega = 7.02$	PASS
3	$B < 0$ (super-exponential growth)	$B = -0.28$	PASS
4	$\lvert C\rvert < 1$ (oscillations don't dominate)	$\lvert C\rvert = 0.014$	PASS
5	$t_c > t_N$ (critical time in future)	$t_c = t_N + 22\text{d}$	PASS
6	$t_c < t_N + \Delta t_{\max}$ (not too far ahead)	$t_c = t_N + 22\text{d}$	PASS
7	$\frac{m\lvert B\rvert\omega}{2\pi} > \lvert C\rvert$ (damping)	$0.23 > 0.01$	PASS
8	$R^2 > 0.80$ (goodness-of-fit)	$R^2 = 0.981$	PASS
	Overall	**8/8**	**ALL PASS**

Decision

- All 8 filters pass $\Rightarrow$ **valid LPPL bubble signal**
- Proceed to bootstrap CI and risk management

Note

- In practice, require $\geq 7/8$ filters
- A single borderline failure (e.g., $\omega = 3.9$) may be acceptable with other strong signals

Step 3: Filter Conditions — BTC 2021

LPPL Filter Conditions: 7/8 Passed

✓	$0.1 \leq m \leq 0.9$	m = 0.900
✓	$4 \leq \omega \leq 25$	ω = 5.35
✓	$B < 0$	B = -0.0037
✓	$\lvert C\rvert < 1$	|C| = 0.0036
✓	$t_c > t_{last}$	tc = 441 > 440
✓	$t_c < t_{last} + \Delta t$	tc = 441 < 500
✗	Damping: $m\lvert B\rvert\omega/(2\pi) > \lvert C\rvert$	0.0028 > 0.0036
✓	$R^2 > 0.80$	R² = 0.9704

7/8 conditions passed → Signal requires caution

TSA_ch13_filter_conditions

Step 4: Bootstrap Confidence Intervals

Block Bootstrap Procedure

1. Fit LPPL on original data $\Rightarrow \hat{\theta}$
2. Compute residuals: $e_i = \ln P_i - \widehat{\ln P_i}$
3. Block size $b = \lfloor \sqrt{88} \rfloor = 9$
4. For each of $B = 500$ replications:
   - Resample residual blocks
   - Create synthetic: $P_i^* = e^{\hat{P}_i + e_i^*}$
   - Re-fit LPPL, apply 8 filters
   - Store $t_c^{(k)}$ if valid
5. CI: $[t_c^{(2.5\%)}, t_c^{(97.5\%)}]$

Bootstrap Results (500 replications)

Parameter	95% CI	Width
t_c (days)	[98, 122]	24d
m	[0.55, 0.90]	0.35
ω	[5.8, 8.4]	2.6
λ	[2.1, 2.9]	0.8

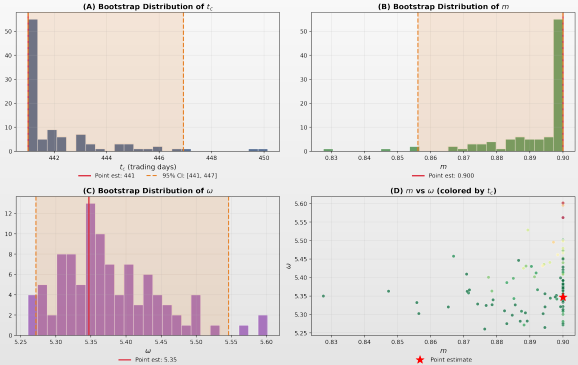
- Lower bound: Oct 26, 2021
- Point estimate: Nov 8, 2021
- Upper bound: Nov 19, 2021
- **Actual peak: Nov 10** $\in$ CI ✓

Quality Check

- Valid bootstrap rate: 412/500 (82.4%)
- Unimodal t_c distribution ✓
- No secondary mode ✓

TSA_ch13_btc2021_bootstrap

Step 4: Bootstrap Parameter Distributions



(A) t_c distribution with 95% CI (B) m distribution (C) ω distribution (D) m vs ω colored by t_c

TSA_ch13_btc2021_bootstrap

Step 5: Multi-Window Confidence Indicator

Window Scan (as of Oct 15)

- 120 overlapping windows: Jul 1–Sep 15 (step 2d), end Oct 15, min 30d

Results

- Passing all filters: **87**/120 ⇒ **CI** = **0.725** Strong bubble signal

t_c Distribution

- Median t_c: Nov 8, 2021; IQR: [Oct 30 – Nov 18]; consistent with bootstrap CI

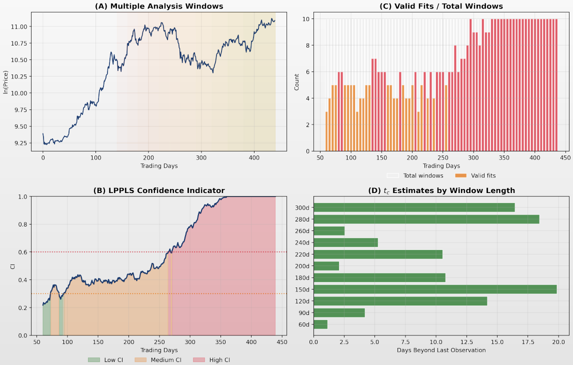
CI Evolution Over Time

Date	CI	Signal
Sep 1	0.15	Normal
Sep 15	0.32	Elevated
Oct 1	0.51	High
Oct 8	0.63	High
Oct 15	0.73	Strong
Oct 22	0.81	Critical
Nov 1	0.78	Critical
Nov 8	0.69	Declining

Key Observation

- CI peaked on Oct 22 (**19 days before peak**) and started declining
- The bubble was already "mature" and unstable

Step 5: Multi-Window CI Dashboard



(A) Multiple windows (B) LPPLS CI with traffic-light zones (C) Valid fits per scan (D) t_c by window length

TSA_ch13_confidence_indicator

Step 6: Risk Management Decision

Signal Summary (Oct 15, 2021)

CI:	**0.725** (strong bubble)
t_c:	Nov 8 $\pm$ 12 days
Filters:	8/8 passed
Bootstrap:	Unimodal, tight

- **Risk-management response should be considered within the estimated critical window**

Heuristic Position Sizing

- Teaching heuristic: target equity $= 1 - \text{CI}$:
  - CI $= 0.725 \Rightarrow$ target $= 27.5\%$
  - **Not an optimal rule** — must be adapted to risk tolerance, transaction costs, and portfolio constraints

Hedging Alternatives

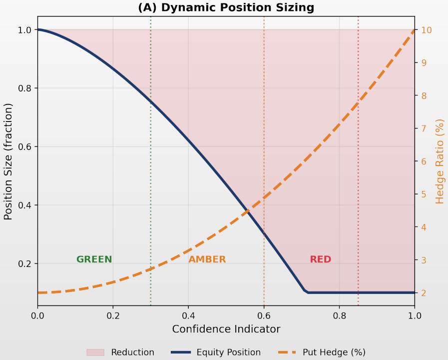
1. **Sell 72.5%** of BTC holdings
2. **Buy puts:** Dec 2021 strike at -20% OTM
3. **Trailing stop:** Set at -15% from peak
4. **Collar:** Sell calls + buy puts (zero cost)

Outcome (Post-Mortem)

Strategy	Max DD	Saved
No action	-77%	—
CI-based sizing	-21%	56pp
Puts at -20%	-20%	57pp
Trailing stop	-15%	62pp

- In this illustrative ex-post backtest, all strategies reduced drawdown relative to buy-and-hold

Step 6: Dynamic Position Sizing



(B) Risk-Management Decision Matrix

$CI < 0.3$	**Normal**	Full position, no hedging
$0.3 \le CI < 0.6$	**Elevated**	Reduce 20%, buy puts (2% AUM)
$0.6 \le CI < 0.85$	**High**	Reduce 50%, increase puts (5%)
$CI \ge 0.85$	**Critical**	Reduce 80%, max puts (8-10%)

(A) Position sizing as function of CI with put hedge overlay (B) Decision matrix: CI thresholds → actions

TSA_ch13_risk_management

Step 7: Post-Mortem and Lessons Learned

Timeline Reconstruction

Jul 20	Window start (BTC at \$29.8k)
Sep 15	CI crosses 0.3 (warning)
Oct 1	CI crosses 0.5 (action level)
Oct 15	CI = 0.73 (reduce exposure)
Oct 22	CI peaks at 0.81
Nov 10	BTC peaks at \$68.8k
Nov 11	Crash begins (−8% day 1)
Dec 31	BTC at \$46.3k (−33%)
Nov 22	BTC at \$15.5k (−77%)

What Went Right

- In this pseudo-real-time reconstruction, the signal was available 26 days before the peak
- t_c estimate within 2 days
- All 8 filter conditions passed for this bubble
- CI trajectory showed clear escalation

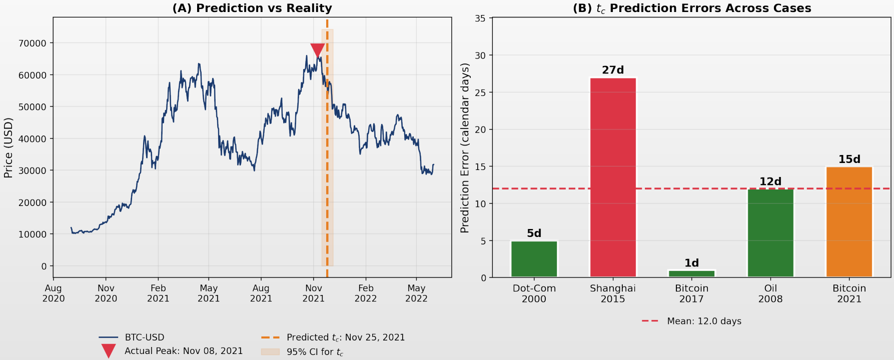
Challenges Encountered

- BTC had **two peaks** in 2021 (Apr + Nov) — multi-scale analysis needed
- Some windows fit the April peak instead
- Transaction costs reduce hedging gains
- Emotional difficulty: reducing position while price rises

Key Lessons

1. LPPL gives a **window**, not a date
2. Act on CI > 0.5, don't wait for CI > 0.8
3. **Gradual** position reduction beats binary
4. Combine with stops for tail protection
5. Document and backtest systematically

Step 7: Post-Mortem — Prediction vs Reality



(A) Predicted t_c vs actual peak with 95% CI (B) t_c prediction errors across all case studies

TSA_ch13_btc2021_data

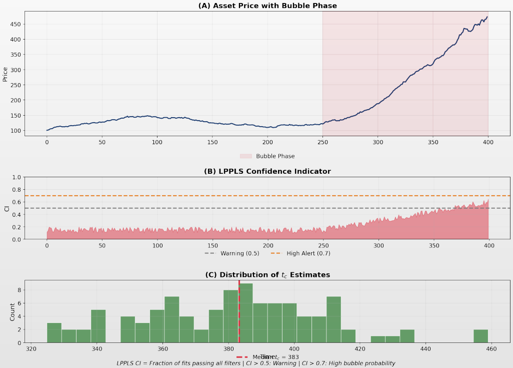
Part VII: LPPLS Confidence Indicator

From Fitting to Monitoring

Real-Time Bubble-Regime Diagnostics

Key Reference: [Sornette et al., 2015]

LPPLS Confidence Indicator: Concept



TSA_ch13_confidence_indicator

Building the Confidence Indicator

The Multi-Scale Approach

1. Fit LPPL on **many windows** (vary start/end)
2. Keep only fits passing **all 8 filters**
3. Count fraction of valid fits → CI

Window Grid

- $t_1 \in [t - 750, t - 30]$ (start dates)
- $t_2 = t$ (end at current date)
- Window range: 30–750 days
- Grid: $\sim$100 windows per day

Confidence Indicator Definition

$$\text{CI}(t) = \frac{\#\{\text{valid LPPL fits at } t\}}{\#\{\text{all fitted windows at } t\}}$$

- $\text{CI} < 0.3$: No LPPL structure detected
- $\text{CI} \in [0.3, 0.5]$: Elevated — monitor
- $\text{CI} \in [0.5, 0.7]$: Strong LPPL structure
- $\text{CI} > 0.7$: Robust bubble-regime signal

Interpretation

- CI is **not** a calibrated crash probability — it is a model-consensus score
- A **rising** CI is more informative than a single high value

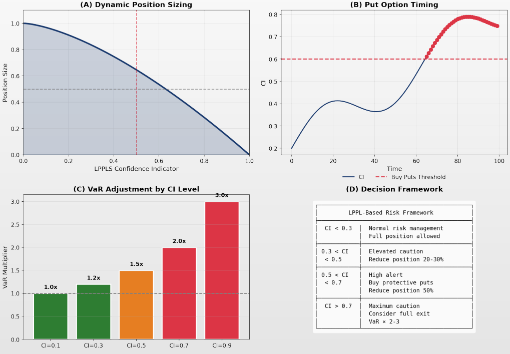
Part VIII: Risk Management

Practical Applications

From Theory to Risk Management

"In theory, there is no difference between theory and practice.
In practice, there is."
— Yogi Berra

Risk Management Applications



LPPL-based risk management: Adjust position, hedging, and VaR based on bubble indicators

TSA_ch13_risk_management

LPPL in Risk Management: Roles and Limits

What LPPL Can Support

- **Early-warning indicator:** flag elevated fragility
- **Stress-testing trigger:** when to run tail scenarios
- **Scenario-analysis input:** inform crash probability
- **Exposure-review tool:** prompt portfolio audit

What LPPL Is **Not**

- Not a mechanical trading rule
- Not a substitute for portfolio optimization
- Not reliable enough for autonomous execution

Heuristic Position Sizing

- Teaching example: $w(t) = w_{\max} \cdot (1 - \mathrm{CI}(t))^{\gamma}$
- $\mathrm{CI} = 0.2 \Rightarrow \text{Position} = 80\%$
- $\mathrm{CI} = 0.5 \Rightarrow \text{Position} = 50\%$
- $\mathrm{CI} = 0.8 \Rightarrow \text{Position} = 20\%$
- **Teaching heuristic only.** Not an optimal allocation rule

From LPPL Signal to Risk-Management Decision

Inputs to the Decision

1. LPPL signal strength (CI level)
2. Uncertainty in t_c (bootstrap CI width)
3. Portfolio exposure and leverage
4. Market liquidity and transaction costs
5. Risk limits and mandate constraints
6. Available hedging instruments
7. Drawdown tolerance and false-alarm cost

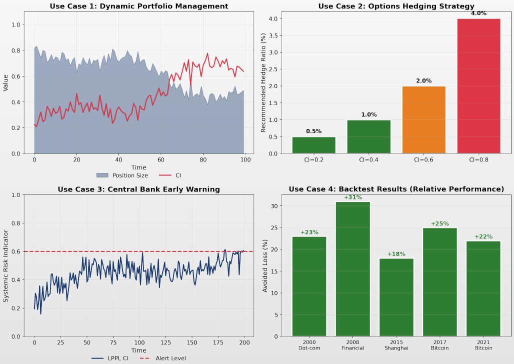
Diagnosis $\neq$ Decision

- LPPL identifies a **regime** — it does not prescribe a trade
- Signal-to-action mapping is **not automatic**
- Same CI $\rightarrow$ opposite actions depending on mandate

Decision Workflow

1. Elevated CI $\Rightarrow$ trigger risk review
2. Assess exposure, hedging cost, false-alarm cost
3. Choose response: reduce, hedge, or monitor
4. Document rationale regardless of outcome

Use Cases: Summary



TSA_ch13_use_cases

Part IX: Modern Extensions

Modern Extensions of LPPL

Beyond the Classical Framework

“The most exciting phrase in science is not ‘Eureka!’ but ‘That’s funny...’ ”
— Isaac Asimov

Multi-Scale LPPLS (MS-LPPLS)

Key Idea [Demos & Sornette, 2017]

- Fit LPPL across **multiple windows** at different scales:
  - **Short** (< 90d): local exuberance
  - **Medium** (90–365d): developing bubbles
  - **Long** (> 365d): macro regime shifts

Advantages

- Captures nested bubbles (bubble within bubble)
- Reduces false positives via multi-scale agreement
- More robust t_c estimation

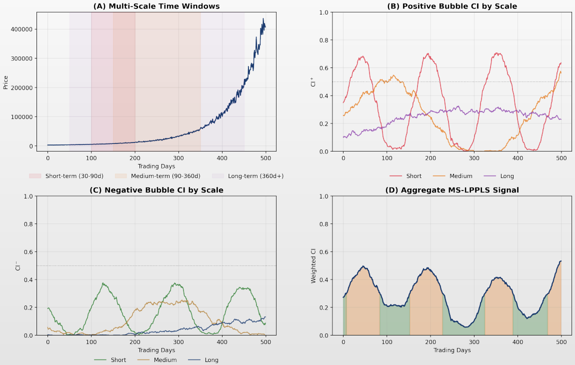
Multi-Scale Confidence Indicator

- Aggregate across scales $s \in \{\text{short, medium, long}\}$:
  - $\mathrm{CI}_{\mathrm{MS}}(t) = \frac{1}{3}\sum_s \mathrm{CI}_s(t)$
  - Crash warning requires **alignment** across ≥ 2 scales

Implementation

1. Window grid: $t_1 \in [t - 750, t - 30]$
2. Fit LPPL on each (t_1, t_2) pair
3. Apply filter conditions per window
4. Cluster qualifying fits by scale
5. Compute scale-specific CI

Multi-Scale LPPLS: Dashboard



(A) Multi-scale time windows (B) Positive bubble CI by scale (C) Negative bubble CI (D) Aggregate signal

TSA_ch13_multiscale

Anti-Bubbles and Negative Bubbles

Definition [Sornette & Zhou, 2002]

- **Anti-bubble**: super-exponential **decline** with log-periodic oscillations, ending in a **rally**
- $\ln P(t) = A + B'(t - t_c')^m[1 + C' \cos(\omega \ln(t - t_c') + \phi')]$
- $t > t_c'$ (post-critical), $B' > 0$ (declining toward A)

Key Difference

- Bubble: $t < t_c$, price accelerates **up**
- Anti-bubble: $t > t_c'$, price decelerates **down**
- Both show log-periodic oscillations

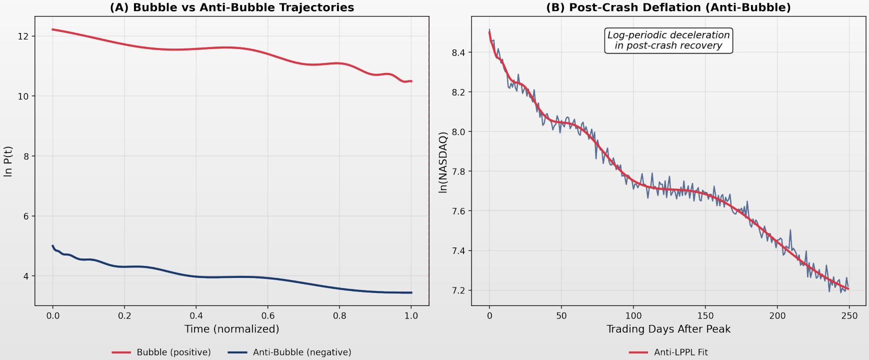
Historical Examples

- **Japan 1990–2003:** Nikkei anti-bubble after 1989 peak, lasting >13 years
- **S&P 500 2000–2003:** Post-dot-com decline with clear log-periodicity
- **Oil 2008–2009:** \$147 $\rightarrow$ \$32, anti-bubble pattern

Practical Implications

- Detect **capitulation points** (end of bear markets)
- Signal regime transitions to recovery
- Symmetric framework: same code, flipped sign

Anti-Bubbles: Visual Evidence



(A) Positive bubble vs negative anti-bubble trajectories (B) Post-crash log-periodic deflation

TSA_ch13_anti_bubble

Bayesian LPPL Estimation

Motivation

- Classical LPPL uses point estimates; **Bayesian LPPL** [Brée & Joseph, 2013; Geraskin & Fantazzini, 2013] provides:
  - Full posterior $p(t_c, m, \omega \mid \text{data})$
  - Credible intervals (not bootstrap)
  - Model comparison via Bayes factors

Prior Specification

- Weakly informative priors:
  - $m \sim \text{Uniform}(0.1, 0.9)$
  - $\omega \sim \text{Uniform}(4, 25)$
  - $t_c \sim \text{Uniform}(t_N, t_N + 0.5T)$
  - $\sigma \sim \text{Half-Cauchy}(0, 1)$

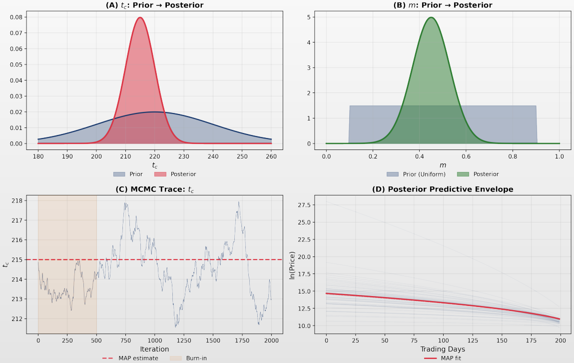
MCMC Implementation (HMC)

1. Define likelihood: $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$
2. Set priors on $(t_c, m, \omega, A, B, C_1, C_2)$
3. Run HMC with 4 chains, 2000 warmup
4. Check convergence ($\hat{R} < 1.01$)

Advantages over Bootstrap

- Incorporates prior knowledge
- Handles multimodality naturally
- Provides model evidence for comparison
- Tools: `PyMC`, `Stan`, `NumPyro`

Bayesian LPPL: Posterior Analysis



(A) t_c prior vs posterior (B) m prior vs posterior (C) MCMC trace (D) Posterior predictive envelope

TSA_ch13_bayesian

Deep Learning for Bubble Detection

Neural Network Approaches

- **LSTM / Transformer:** Temporal pre-crash dynamics [Chatzis et al., 2018]
- **CNN on returns:** Price windows as images
- **Hybrid LPPL-NN:** LPPL features (m, ω, CI) as inputs

Training Strategy

1. Label historical windows: bubble vs. normal
2. Features: returns, volatility, LPPL residuals
3. Train on pre-2010, validate on 2010–2020
4. Binary output: $P(\text{crash in } K \text{ days})$

Comparison: LPPL vs. Deep Learning

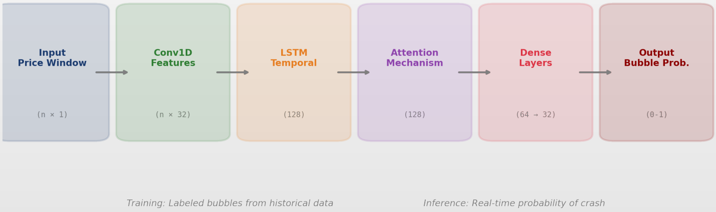
Criterion	LPPL	DL
Interpretability	High	Low
Theory-driven	Yes	No
Data requirement	Low	High
Flexibility	Low	High
False positive rate	Medium	Variable
Generalization	Good	Risk

Best Practice

- **Ensemble:** Combine LPPL theory + DL pattern recognition
- Use LPPL CI as a feature, not a replacement
- DL can improve pattern detection but does not remove data-snooping or false-positive risk

Deep Learning: CNN-LSTM Architecture

Deep Learning for Bubble Detection: CNN-LSTM Architecture



TSA_ch13_deep_learning

Regime-Switching LPPL

Framework [Ardila-Alvarez et al., 2021]

- LPPL + **Markov-Switching**; $S_t \in \{\text{Normal}, \text{Bubble}, \text{Post-crash}\}$

$$\ln P_t = \begin{cases} \mu + \sigma \varepsilon_t & S_t = \text{Normal} \\ \text{LPPL}(t) + \sigma_b \varepsilon_t & S_t = \text{Bubble} \\ \alpha + \beta t + \sigma_c \varepsilon_t & S_t = \text{Post} \end{cases}$$

Transition Matrix

$$P = \begin{pmatrix} p_{NN} & p_{NB} & 0 \\ 0 & p_{BB} & p_{BP} \\ p_{PN} & 0 & p_{PP} \end{pmatrix}$$

- No direct Normal $\rightarrow$ Post-crash transition

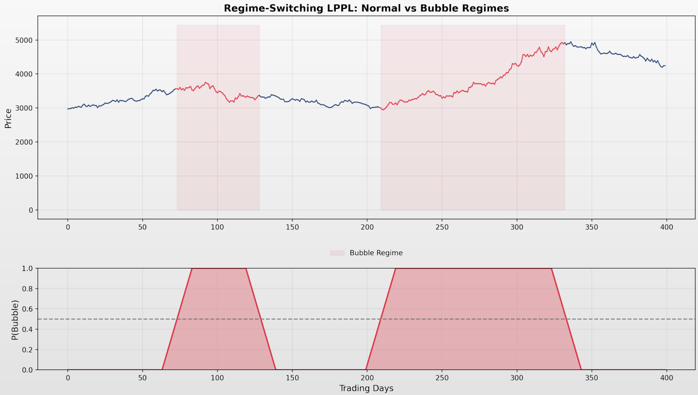
Key Innovation

- **Endogenous regime detection:** No need to pre-specify bubble windows
- **Filtered probabilities:** $P(S_t = \text{Bubble} \mid \mathcal{F}_t)$ in real time
- **Duration modeling:** Expected time in bubble state

Estimation

1. EM algorithm for regime parameters
2. Nested LPPL optimization in bubble state
3. Hamilton filter for state probabilities
4. AIC/BIC for number of regimes

Regime-Switching LPPL: Visualization



TSA_ch13_regime_switching

LPPL with Sentiment and Alternative Data

Sentiment-Enhanced LPPL [Hüsler et al., 2013]

- **Social media:** Twitter/X sentiment, Reddit (r/wallstreetbets)
- **News tone:** NLP-based bullishness index
- **Search trends:** Google Trends for "buy [asset]"
- **On-chain data:** Wallet activity, exchange inflows

Extended Model

- Hazard rate: $h(t) = h_0(t)[1 + \gamma \cdot \text{Sent}(t)]$
- Higher sentiment $\Rightarrow$ higher crash hazard

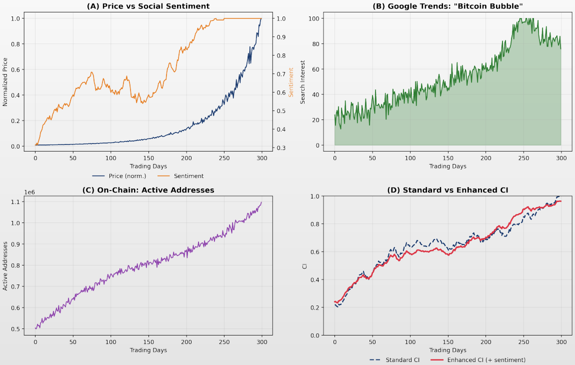
Evidence

- Social media volume spikes **precede** crashes by 1–4 weeks [Preis et al., 2013]
- Google "Bitcoin bubble" search interest rose sharply before Dec 2017 crash
- Combining CI with sentiment indicators may help reduce false positives, but performance is dataset-dependent

Crypto-Specific Indicators

- **NVT ratio:** Network value to transactions
- **MVRV:** Market value vs. realized value
- **Exchange inflows:** Selling pressure proxy
- Combined with LPPL CI for multi-factor alert

Sentiment-Enhanced LPPL



(A) Price vs social sentiment (B) Google Trends (C) On-chain metrics (D) Standard vs enhanced CI

TSA_ch13_sentiment

Real-Time Monitoring: The FCO System

Financial Crisis Observatory [ETH Zurich]

- Operational system monitoring a **large asset universe** daily since 2014:
  - Equities (indices, sectors, single stocks)
  - Fixed income (sovereign, corporate)
  - Commodities (energy, metals, agriculture)
  - Cryptocurrencies (BTC, ETH, top 50)
  - Real estate indices

Architecture

1. Daily data ingestion (API feeds)
2. Multi-scale LPPL fitting (GPU cluster)
3. Filter & classify qualified fits
4. Compute CI per asset and scale
5. Generate dashboard and alerts

Dashboard Output

- For each asset, the FCO reports:
  - **Bubble status:** positive / negative / neutral
  - **CI value:** $[0, 1]$ per time scale
  - t_c **distribution:** When the regime may end
  - **Traffic light:** Green / Amber / Red

Track Record

- Flagged Bitcoin 2017, 2021 bubbles early
- Detected Shanghai 2015 bubble at medium scale
- Published monthly reports since 2014
- Exact performance metrics depend on asset class and threshold

Extensions: Summary

Extension	Key Innovation	Best For
Multi-Scale LPPLS	Multiple time horizons	Nested bubbles, robust CI
Anti-Bubbles	Symmetric decline model	Bear market reversals
Bayesian LPPL	Full posterior on t_c	Uncertainty quantification
Deep Learning	Data-driven patterns	Large datasets, ensembles
Regime-Switching	Endogenous state detection	Automated monitoring
Sentiment + LPPL	Behavioral covariates	Crypto, social-driven markets
FCO System	Production-scale pipeline	Institutional risk management

- **Common theme:** All extensions preserve the core LPPL insight (super-exponential growth + log-periodicity) while addressing specific limitations of the classical framework

Part X: Limitations

Critiques and Limitations

Honest Assessment of LPPL Models

"All models are wrong, but some are useful."
— George Box

Academic Critiques

Statistical Concerns [Bree et al., 2013]

1. **Overfitting:** 7 parameters for ∼100 points
2. **Look-ahead bias:** easy to fit ex-post
3. **Selection bias:** only successes reported
4. **Multiple testing:** many windows, some will fit
5. **Data snooping:** parameters tuned to known crashes

Response to Critiques

- Strict filters (all 8) + out-of-sample evaluation
- Report CIs, false-positive rates, placebo tests

Empirical Concerns

1. False positives occur
2. Timing uncertainty is large
3. Not all bubbles show LPPL patterns
4. Not all LPPL fits lead to crashes
5. Parameters vary across markets

The Honest Assessment

- LPPL is a **fever thermometer**: elevated signal ⇒ elevated risk
- Does not guarantee crash; use as one input among many
- Humility is essential

When LPPL Signals Fail

Failure Modes

1. Market enters **sideways correction** instead of crashing
2. Estimated t_c **shifts forward** as new data arrive
3. Bubble **deflates gradually** (no sharp crash)
4. Signal appears only for **narrow windows**
5. Log-periodicity may be a **fitting artefact**
6. High R^2 does not prove **predictive validity**

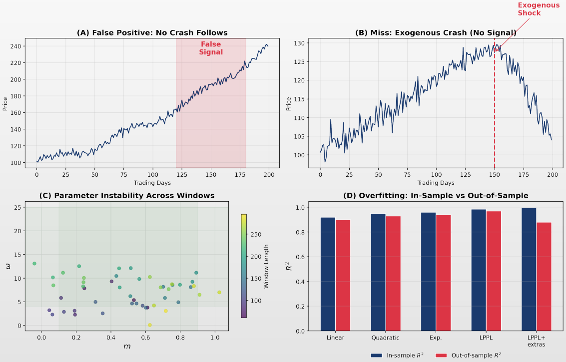
Historical False Positives

- US equities 2014–2015: LPPL signals, no crash
- Bitcoin 2019: brief LPPL signal, gradual decline
- European debt 2011: exogenous policy, not LPPL bubble

Honest Interpretation

- A false positive may still indicate overheating or fragility
- But it **must be reported honestly**
- Never hide failures behind successful case studies

When LPPL Fails: Visual Evidence



(A) False positive: signal without crash (B) Miss: exogenous shock (C) Parameter instability (D) Overfitting risk

TSA_ch13_limitations

Ambiguous Signals: When Warnings Do Not Become Crashes

How LPPL Warnings Can Be "Right" But Not Useful

1. Market enters a **sideways correction** instead of crashing
2. Estimated t_c **moves forward** as new data arrive ("rolling t_c")
3. Bubble **deflates gradually** rather than bursting (slow air leak)
4. Only a **narrow subset** of windows passes the filters
5. Apparent oscillations **disappear** under placebo testing
6. Signal is valid as **fragility evidence** but invalid as a crash-timing signal

Example: S&P 500, 2014–2015

- LPPLS CI rose above 0.5 in late 2014
- Market continued rising for 6 months
- August 2015: -12% correction (not a crash)
- LPPL signal was consistent with overheating but timing was off by months

Key Message

- A false positive is not necessarily useless: it may indicate overheating or fragility
- But it must be **reported honestly** and separated from successful warnings
- The scientific question is not "Was there a crash?" but "Was the regime consistent with endogenous instability?"

What LPPL Can and Cannot Do

LPPL **Can**

- Detect super-exponential price acceleration
- Estimate the most likely end-of-bubble regime
- Provide probabilistic risk windows
- Quantify log-periodic oscillations
- Serve as one input in a multi-indicator framework
- Identify non-bubble patterns (negative control)

LPPL **Cannot**

- Determine the exact date or magnitude of a crash
- Guarantee that a regime break will occur
- Detect exogenous shocks (pandemics, wars)
- Work reliably on short windows (< 100 days)
- Replace fundamental analysis or due diligence
- Distinguish bubbles from genuine regime shifts

The Bottom Line

- LPPL is a **diagnostic tool**, not a crystal ball
- Identifies a statistical pattern consistent with endogenous bubble dynamics
- A positive signal indicates elevated instability; it does not make a crash certain

Master Assignment: LPPL Bubble Diagnostic

Task

- Choose one asset and one historical period (bubble or non-bubble)
- Perform a complete LPPL diagnostic

Required Steps

1. Download daily prices; transform to log-prices
2. Fit exponential, quadratic, and LPPL models
3. Estimate LPPL via partial linearization + DE
4. Apply the 8 filter conditions
5. Bootstrap CIs for t_c ($N \geq 200$)
6. LPPLS CI across ≥ 3 nested windows
7. Compare against benchmark models

Deliverable

- One-page interpretation covering:
  - Signal strength and parameter plausibility
  - Uncertainty (CIs, window sensitivity)
  - Benchmark comparison results
  - False-positive risk assessment
  - Risk-management implications

Grading Criteria

- Correct estimation: 25%
- Robustness and benchmark comparison: 25%
- Uncertainty and false-positive analysis: 25%
- Critical interpretation and reproducibility: 25%

Reproducibility Requirement

Submission Deliverables

1. Python notebook or standalone Quantlet
2. Data download script (reproducible)
3. LPPL estimation function
4. Filter-condition function (all 8 checks)
5. Benchmark-model comparison (exp, quad, LPPL)
6. Bootstrap t_c confidence interval
7. LPPLS confidence-indicator plot
8. Negative-control or false-positive test
9. One-page written interpretation

Grading Weights

Component	Weight
Correct LPPL implementation	20%
Benchmark comparison	15%
Bootstrap and uncertainty	15%
Multi-window robustness	15%
Negative control / false-positive	10%
Interpretation and limitations	15%
Reproducibility and code quality	10%

Reproducibility Standard

- Code must run from a clean environment
- All data downloaded programmatically
- Random seeds fixed for reproducibility
- Results must be obtainable by the grader

Summary: Key Takeaways

Theoretical Foundation

1. Bubbles show **super-exponential** growth
2. LPPL interprets bubbles as **herding** + positive feedback
3. Exhibit **log-periodic** oscillations
4. Same physics as **phase transitions**

The LPPL Model

$$\ln P = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi)$$

- 7 parameters (3 linear, 4 nonlinear)
- Partial linearization + DE estimation
- 8 filter conditions for validity
- **Bootstrap CI** for uncertainty

Case Study Results (Mean error: 10.2d)

Case	Error	95% CI
Bitcoin 2021	2d	[478, 485]
Bitcoin 2017	1d	[167, 169]
NASDAQ 2000	9d	[369, 383]
Shanghai 2015	27d	[169, 187]
Oil 2008	12d	[382, 423]
COVID 2020	N/A	No bubble

Key Message

- LPPL + Bootstrap CI = Structured bubble-regime diagnosis with uncertainty bounds

Course Conclusion

The Central Message

- LPPL is a **structured diagnostic framework** for identifying endogenous bubble regimes
- Used responsibly $\Rightarrow$ early-warning systems, stress testing, risk review
- Used mechanically $\Rightarrow$ overfitting, selection bias, false confidence

The Scientific Question

- **Not**: "Did LPPL predict the crash?"
- **Rather**: "Did LPPL provide robust, benchmark-adjusted evidence of an unstable bubble regime?"

What Students Should Be Able To Do

- **Fit** LPPL via partial linearization + DE
- **Validate** with 8 filters + residual diagnostics
- **Benchmark** against simpler models
- **Stress-test** with bootstrap, placebo, controls
- **Interpret** critically — evidence vs. overfit

Questions? danpele@ase.ro

Acronyms (1/2)

Core Model & Framework

- **LPPL** — Log-Periodic Power Law
- **LPPLS** — Log-Periodic Power Law Singularity
- **CI** — Confidence Indicator
- **EMH** — Efficient Market Hypothesis
- **CSI** — Continuous Scale Invariance
- **DSI** — Discrete Scale Invariance
- **RG** — Renormalization Group
- **FCO** — Financial Crisis Observatory
- **ARCH** — Autoregressive Cond. Heteroskedasticity
- **LM** — Lagrange Multiplier

Estimation & Testing

- **OLS** — Ordinary Least Squares
- **DE** — Differential Evolution
- **SSR** — Sum of Squared Residuals
- **LR** — Likelihood Ratio
- **AIC** — Akaike Information Criterion
- **BIC** — Bayesian Information Criterion
- **IQR** — Interquartile Range
- **RMSE** — Root Mean Square Error
- **MAE** — Mean Absolute Error
- **BDS** — Brock–Dechert–Scheinkman (test)

Acronyms (2/2)

Machine Learning & Bayesian

- **MCMC** — Markov Chain Monte Carlo
- **HMC** — Hamiltonian Monte Carlo
- **LSTM** — Long Short-Term Memory
- **CNN** — Convolutional Neural Network
- **NN** — Neural Network
- **DL** — Deep Learning
- **NLP** — Natural Language Processing
- **MS** — Multi-Scale / Markov-Switching
- **GPU** — Graphics Processing Unit
- **NBER** — National Bureau of Economic Research
- **COVID** — Coronavirus Disease

Finance & Markets

- **BTC** — Bitcoin
- **ETH** — Ethereum
- **USD** — United States Dollar
- **WTI** — West Texas Intermediate
- **S&P** — Standard & Poor's
- **NASDAQ** — Natl. Assoc. Securities Dealers Autom. Quotations
- **FOMO** — Fear Of Missing Out
- **HODL** — Hold On for Dear Life
- **ICO** — Initial Coin Offering
- **OTM** — Out of The Money
- **NVT** — Network Value to Transactions
- **MVRV** — Market Value to Realized Value

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